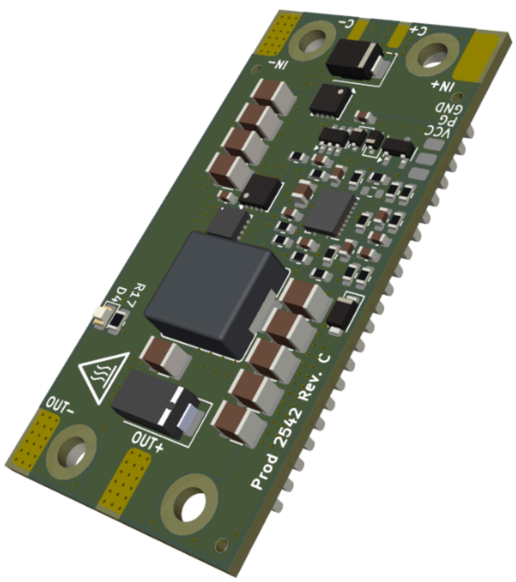
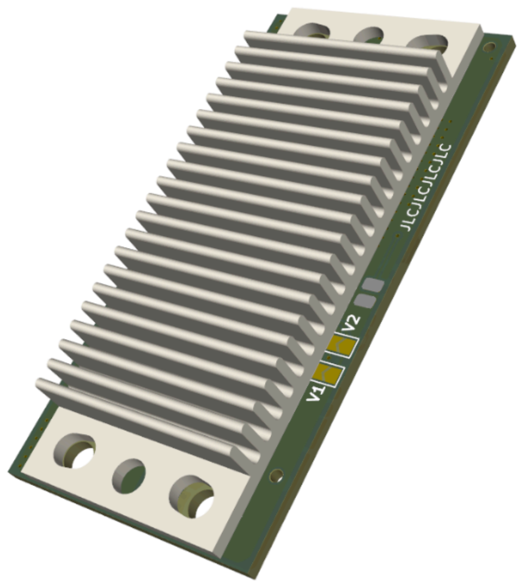
November 2025

*Per Unitatem Ad Astra*

# Super Step Down Documentation – SSDV3 Rev. C




**Description:**

A non-isolated, 50 V input, DC/DC converter module designed for stepping down battery power on unmanned aerial systems (UAS). The SSDV3 is the third major design iteration of UBCO Aerospace's Super Step Down series.

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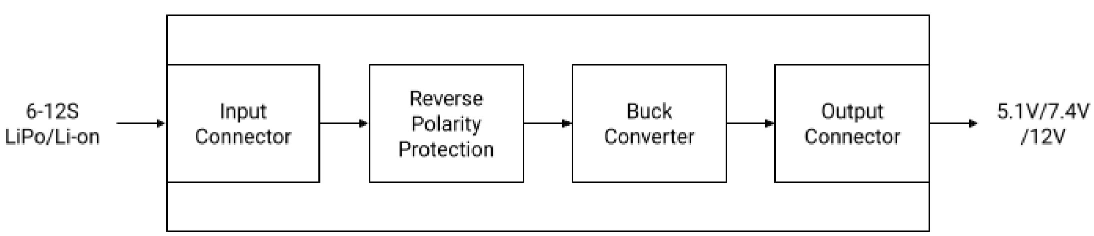
**Team:** UAV Division, Electrical Subteam
**Revision:** Rev. C


**System Diagram:**




**GitHub:** [https://github.com/UBCOAerospaceClub/2025_SSDV3](https://github.com/UBCOAerospaceClub/2025_SSDV3)

## Important Notice

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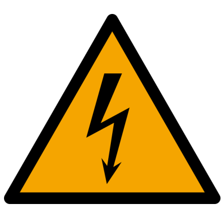
## Safety Precautions

**Warning:**
This device is intended to be directly connected to Li-Ion batteries which, under short-circuit conditions or unsafe handling, have the potential to cause fire or explosion. Take necessary precautions to insulate the device inputs and mitigate the risk of a short circuit.

**Caution:**
This device becomes hot under normal operation. Wait at least two minutes after powering off and allow the device to cool down to suitable temperatures before handling. When servicing, do not use low melting-point solder alloys, such as those that contain bismuth or indium.



**Caution:**
The input voltage to this device can be in excess of 60 VDC. Keep conductive materials away from the exposed device inputs. The device must be powered down before it can be handled or serviced.

**Caution:**
This device is susceptible to damage from electrostatic discharge (ESD). At minimum, refrain from touching any electrical components attached to the PCB when handling without precautionary ESD measures.

## Design Revisions

| Rev. | Date | Description |
|---|---|---|
| A-1 | 2025-07 | Concept design using the LM70880 buck converter and LM5069-1 hot-swap controller. Abandoned due to supply chain issues and cost constraints. |
| A-2 | 2025-08 | Concept design using the LM5146 buck controller and TPS2490 hot-swap controller. Abandoned due to space and cost constraints. |
| A-3 | 2025-09 | Concept design using the LM5146 buck controller in a custom 1/8th brick form factor. Accepted design for revision. |
| B | 2025-10 | Concept design using the cheapest parts available on LCSC. Limited part availability and NRND parts made future production of this revision difficult. |
| C | 2025-10 | Prototype design featuring reverse polarity protection in exchange for a smaller TVS diode on the input. Smaller, active-production 3x3mm PQFN-8 FETs used in place of SO-8 FETs. Used for preliminary integration testing and verification. |

# Table of Contents



# 1 Specifications

## 1.1 Absolute Maximum Ratings

Under typical ambient temperature of $T_A$ = 21 °C, unless noted otherwise.

| Parameter | | Min | Max | Unit |
|---|---|---|---|---|
| $V_{IN}$ | Input voltage from `VIN+` to `VIN-` and `C+` to `C-`* | −60 | 60 | V |
| | Maximum transient input voltage* | −70 | 70 | |
| $I_{IN}$ | DC Input current | 0 | 6 | A |
| $V_{EN}$ | Input voltage from `EN` to `GND`* | −60 | 60 | V |
| $V_{OUT}$ | Output voltage from `VOUT+` to `VOUT-` | 0 | 20 | V |
| $I_{OUT}$ | DC Output current | | 8 | A |
| $P_{OUT}$ | Output power for 60 seconds | | 100 | W |
| | Continuous output power with heatsink | 0 | 60 | |
| | Continuous output power without heatsink | 0 | 40 | |
| $V_{CC}$ | Output voltage from `VCC` to `GND` | 0 | 7.8 | V |
| $I_{VCC}$ | Output current from `VCC` | 0 | 0.02 | A |
| $V_{PG}$ | Output voltage from `PG` to `GND` | 0 | 7.8 | V |
| $T_A$ | Operating ambient temperature range | 0 | 55 | °C |
| $T_{STG}$ | Storage temperature range | 0 | 85 | °C |

*The input electrolytic capacitor is not protected from reverse polarity.

## 1.2 Electrical Characteristics

Under typical ambient temperature of $T_A$ = 21 °C, unless noted otherwise.

| Parameter | | Test Conditions | Min | Typ. | Max | Unit |
|---|---|---|---|---|---|---|
| **1.2.1 Input Characteristics** | | | | | | |
| $V_{IN}$ | Input voltage operating range | | 20 | 50 | 60 | V |
| $I_{Q,ON}$ | Quiescent input current, no load | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1V, no attached load to VOUT | | 71 | | mA |
| $I_{Q,OFF}$ | Quiescent input current, UVLO | $V_{IN}$ = 10 V | | 2 | | mA |
| **1.2.2 Dynamic Characteristics** | | | | | | |
| $f_{SW}$ | Switching frequency | | | 420 | | kHz |
| $t_{SS}$ | Soft-start time | $V_{IN}$ = 50 V | | 4 | | ms |
| $I_{IN,MIN}$ | Minimum startup supply current | $V_{IN}$ = 50 V, $V_{OUT}$ = 12.0 V | | 0.6 | | A |
| $\Delta V_{MAX}$ | Maximum transient output deviation | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1V, 5 A load step | | ±1.5 | ±2.5 | % |
| **1.2.3 Output Characteristics** | | | | | | |
| η | Efficiency | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, $I_{OUT}$ = 4.3 A | | 85 | | % |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 7.4 V, $I_{OUT}$ = 2.7 A | | 87 | | |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 12.0 V, $I_{OUT}$ = 3.3 A | | 91 | | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 5.1 V | | TBD | | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 7.4 V | | TBD | | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 12.0 V | | TBD | | |
| $\Delta V_{PP}$ | Peak-to-peak output voltage ripple | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V | | 9.4 | 15.0 | $mV_{PP}$ |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 7.4 V | | 12.3 | 19.0 | |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 12.0 V | | 27.3 | 35.0 | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 5.1 V | | TBD | | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 7.4 V | | TBD | | |
| | | $V_{IN}$ = 25 V, $V_{OUT}$ = 12.0 V | | TBD | | |

*The input electrolytic capacitor is not protected from reverse polarity.

## 1.2 Electrical Characteristics (continued)

Under typical ambient temperature of $T_A$ = 21 °C, unless noted otherwise.

| Parameter | | Test Conditions | Min | Typ. | Max | Unit |
|---|---|---|---|---|---|---|
| **1.2.4 Protection Features** | | | | | | |
| $I_{OCP}$ | OCP current limit threshold | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V | | 10.2 | | A |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 7.4 V | | 10.5 | | |
| | | $V_{IN}$ = 50 V, $V_{OUT}$ = 12.0 V | | 11.6 | | |
| OCP | Overcurrent protection | $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V | | PASS | | : ) |
| RPP | Reverse polarity protection | $V_{IN}$ = 50 V | | FAIL* | | : ( |
| **1.2.5 Auxiliary IO Characteristics** | | | | | | |
| $V_{EN,ON}$ | Enable voltage | Input voltage rising | | 16.5 | | V |
| $V_{EN,OFF}$ | Undervoltage lockout threshold | Input voltage falling | | 15.0 | | V |
| $V_{VCC}$ | VCC nominal voltage | $V_{IN}$ = 50 V | | 7.5 | | V |
| $V_{PG}$ | PG nominal voltage | $V_{IN}$ = 50 V | 0 | 7.5 | | V |

## 1.3 Performance Curves

Under typical ambient temperature of $T_A$ = 21 °C, unless noted otherwise. Sample size of 1.

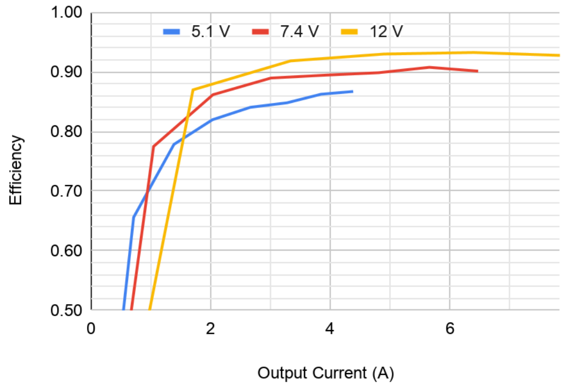



Figure 1.1: Efficiency vs output current, $V_{IN}$ = 50 V

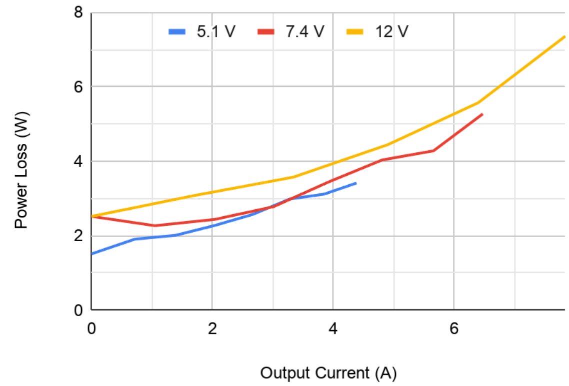



Figure 1.2: Power loss vs output current $V_{IN}$ = 50 V

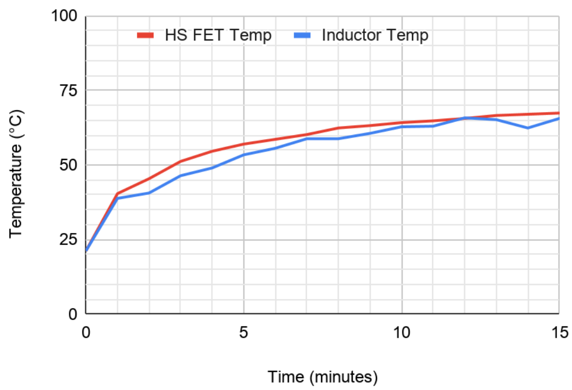



Figure 1.3: Thermal response with heatsink, $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, $I_{OUT}$ = 4.4 A

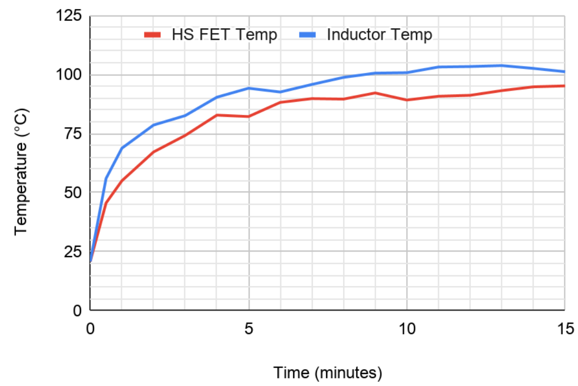



Figure 1.4: Thermal response with heatsink, $V_{IN}$ = 50 V, $V_{OUT}$ = 12 V, $I_{OUT}$ = 6.3 A

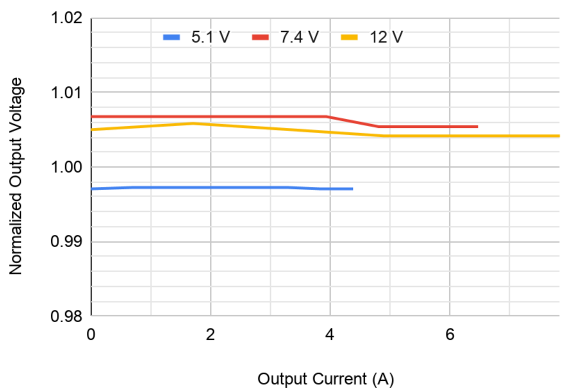



Figure 1.5: Regulation vs output current, $V_{IN}$ = 50 V


Figure 1.6: Output voltage waveform, $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, no load

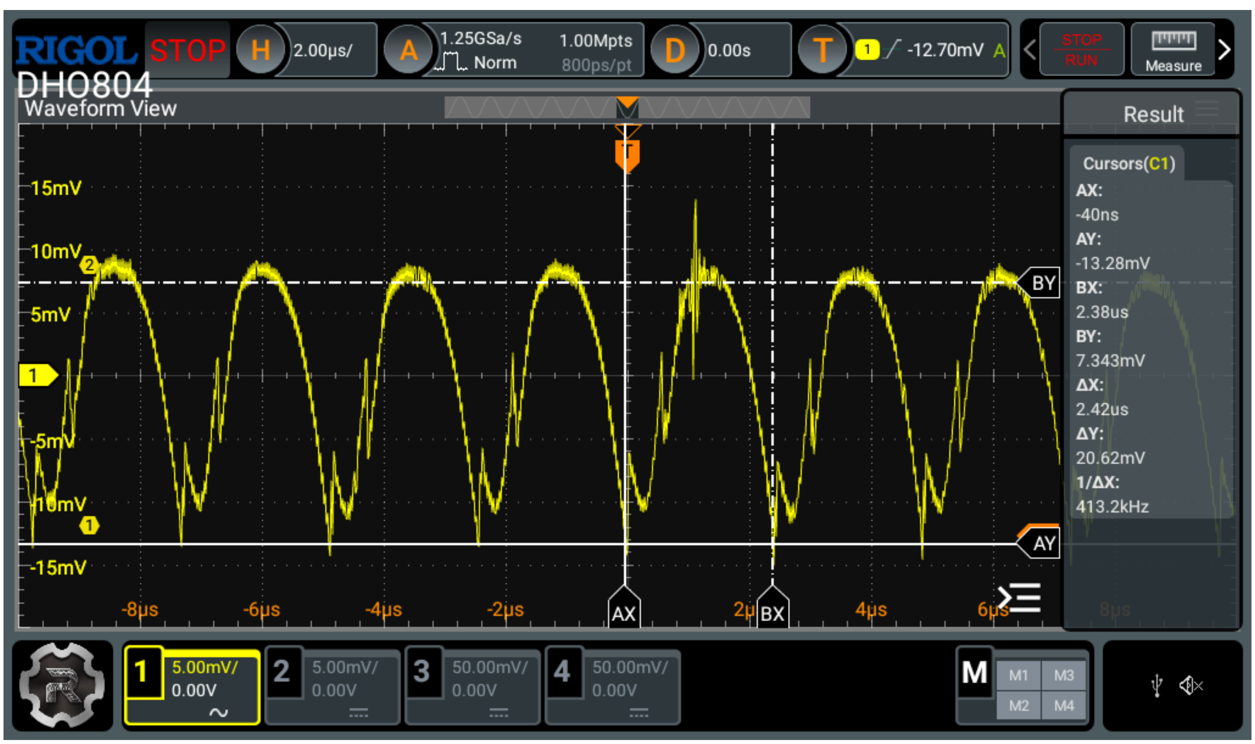



Figure 1.7: Output voltage waveform, $V_{IN}$ = 50 V, $V_{OUT}$ = 12 V, no load

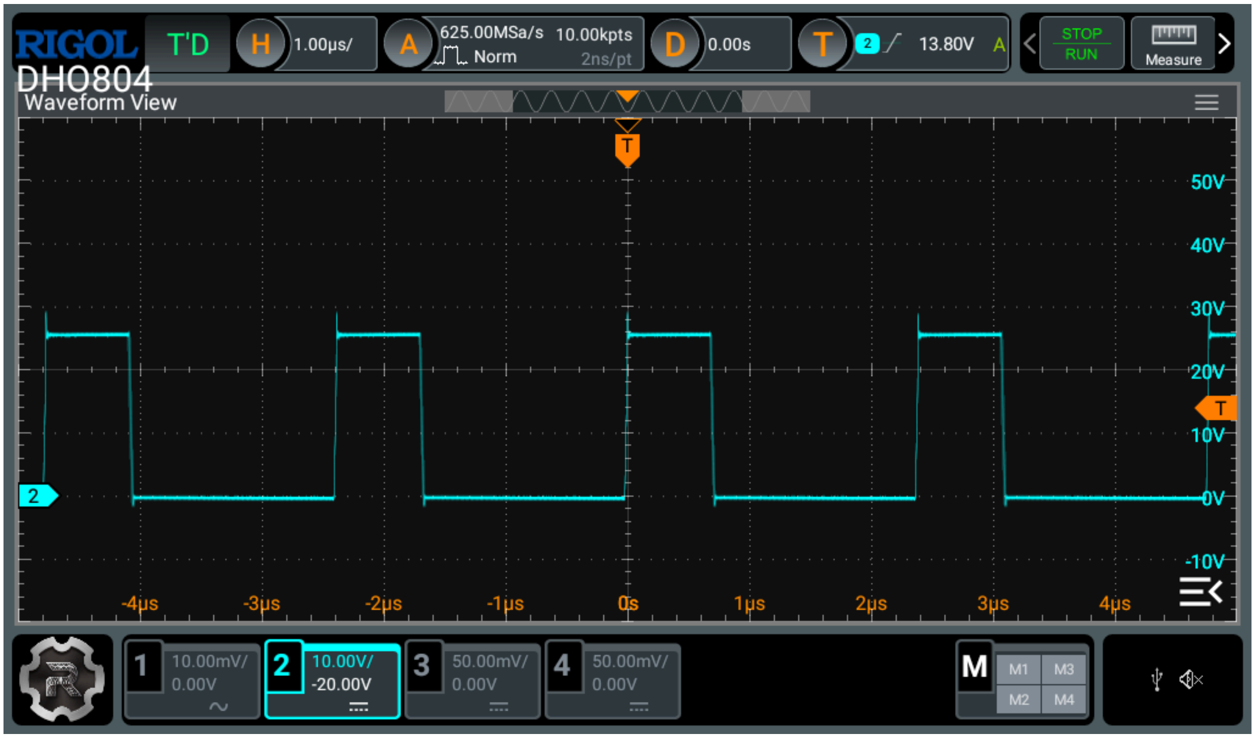



Figure 1.8: Switch-node waveform, $V_{IN}$ = 25 V, $V_{OUT}$ = 5.1 V, no load

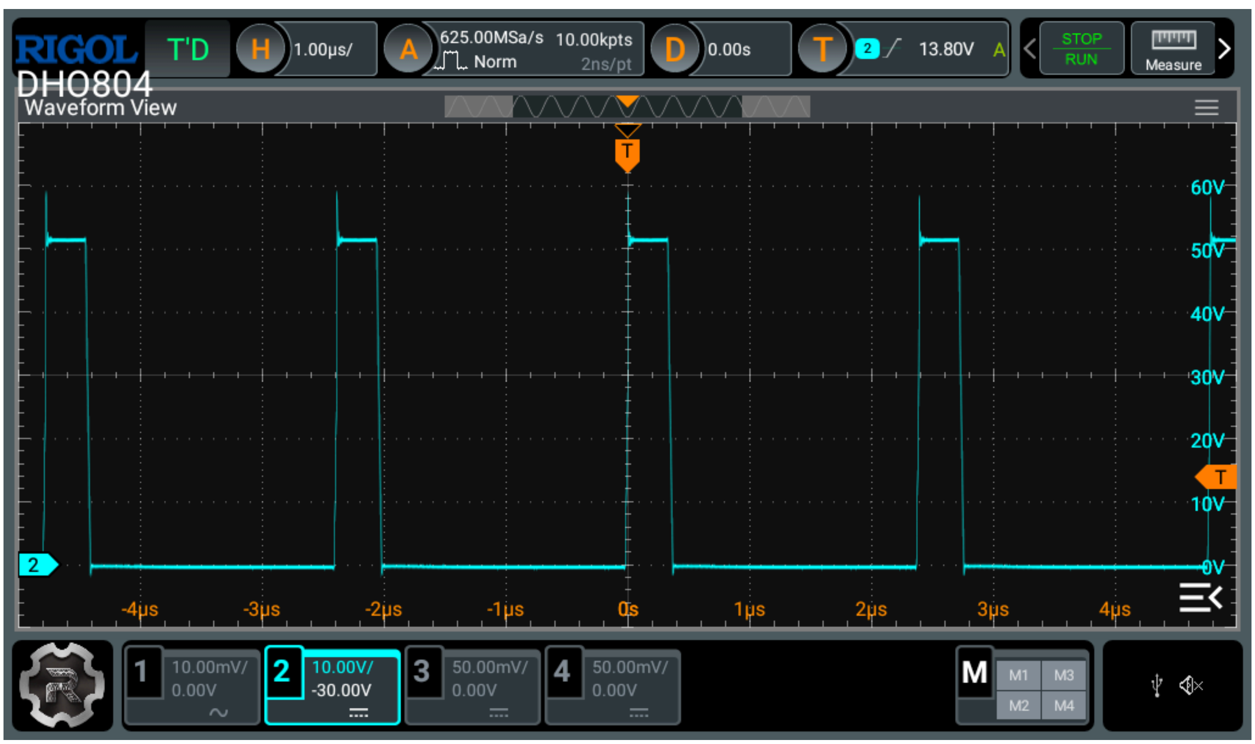



Figure 1.9: Switch-node waveform, $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, no load

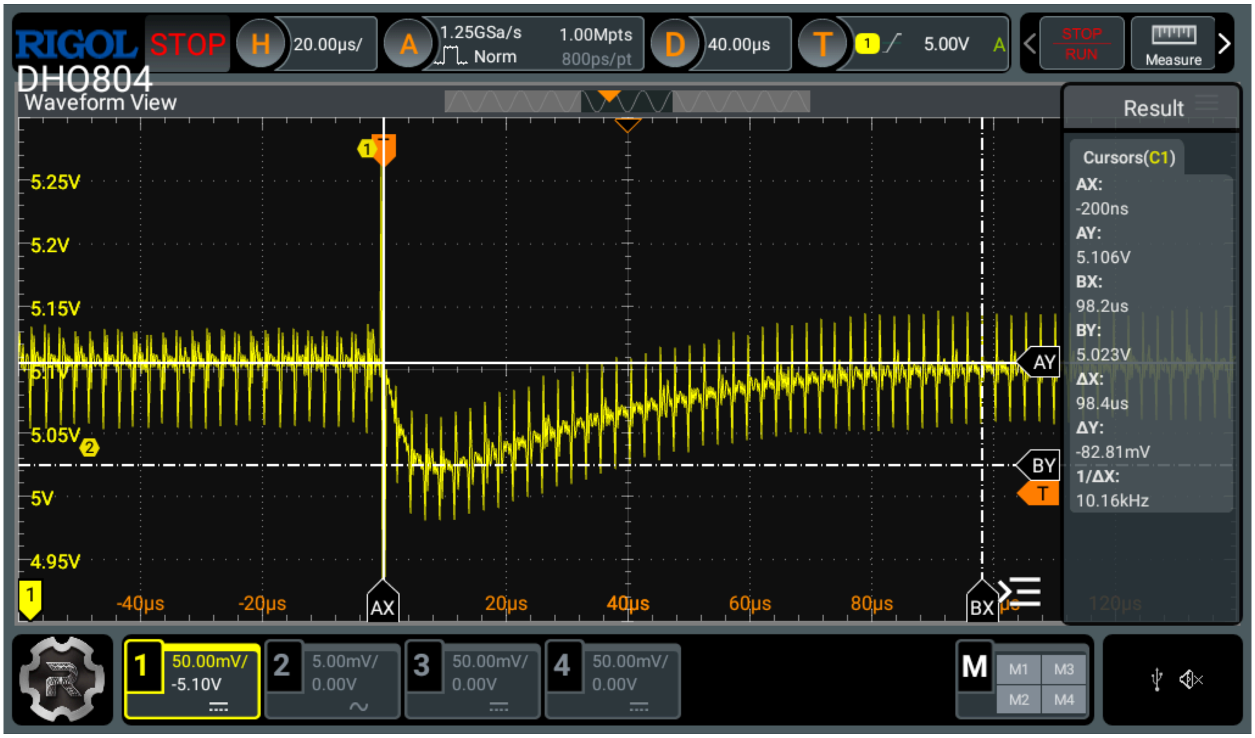



Figure 1.10: Output load-on transient, $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, $I_{OUT}$ = 5 A

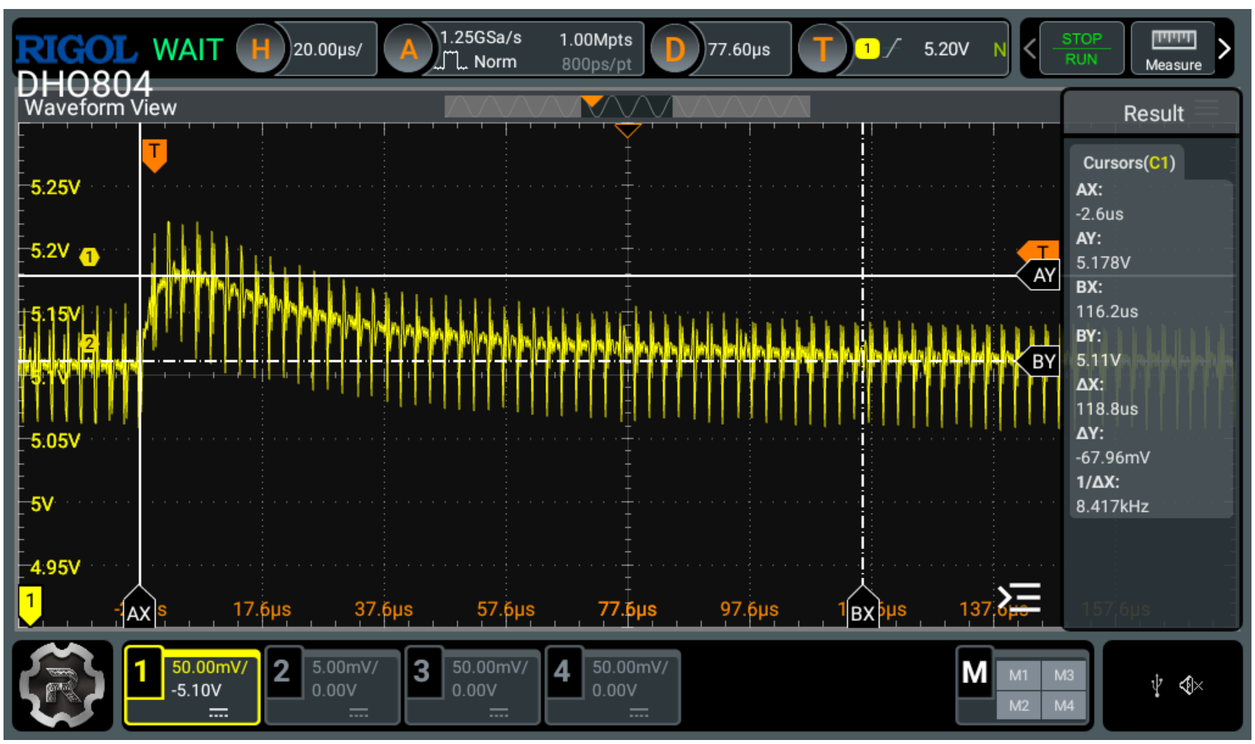



Figure 1.11: Output load-off transient, $V_{IN}$ = 50 V, $V_{OUT}$ = 5.1 V, $I_{OUT}$ = 5 A

# 2 Usage

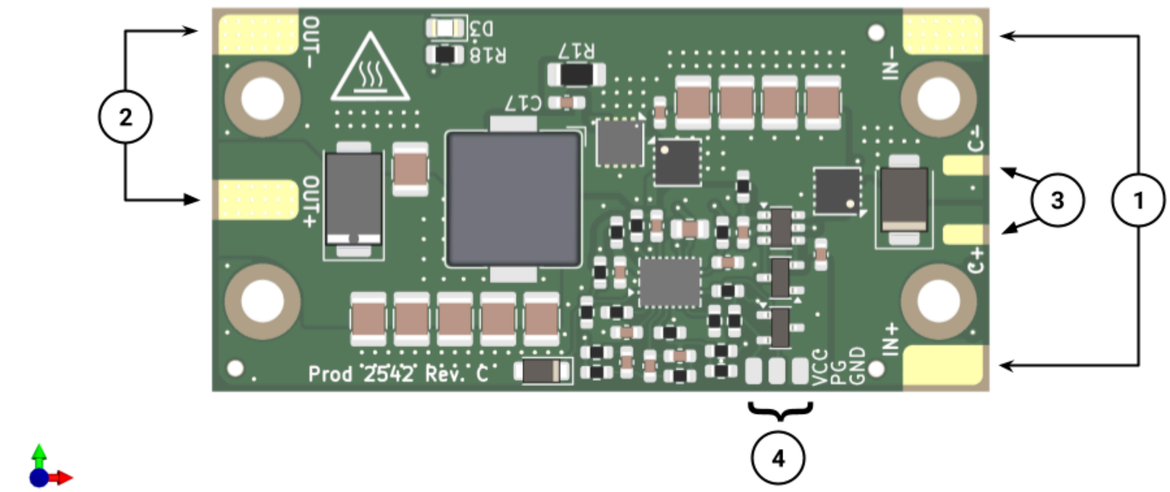
Please read the Safety Precautions before continuing.

## 2.1 Functional Description

The SSDV3 is a non-isolated, wide input voltage step-down module intended to provide a reliable voltage source for electrically demanding systems requiring power from LiPo batteries. The design offers excellent disturbance rejection from battery voltage sag and sudden load spikes, giving much better performance than other hobbyist devices, making it a strong alternative to commercially-available battery eliminator circuits (BECs) while maintaining a small form-factor. Its extremely robust output regulation makes the SSDV3 best suited for supplying low-voltage devices, especially high-performance embedded computers or sensitive electronics that specify strict 5 V power requirements.

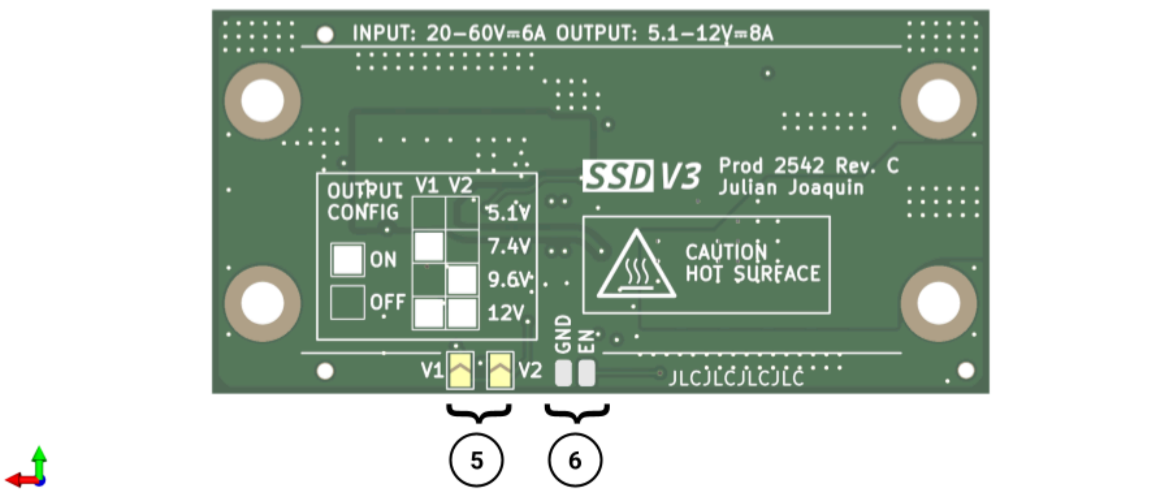
What makes the SSDV3 unique from other market BECs is its design for reliability. The step-down circuitry is protected by a TVS diode to address potential ESD on the battery lines. Copper traces leading from the input pads are designed significantly above specification to mitigate damage from inrush current whenever the battery is plugged in. Additionally, an N-channel MOSFET at the input acts as an ideal diode to protect the module and downstream electronics from a reversed battery connection, at minimal cost to the step-down's efficiency.

## 2.2 Interfaces




1. Power Input Pads VIN
2. Power Output Pads VOUT
3. Input Capacitor Pads C
4. Power Good Indicator Pads VCC, PG, GND

Figure 2.1: Module top interfaces




5. Output Voltage Jumpers V1 and V2
6. Enable Pads EN and GND

Figure 2.2: Module bottom interfaces

## 2.3 Installation Guidelines

- Ensure wires of at least 18 AWG have been soldered onto the input power and output power pads before installing the module. Use power connectors rated for the intended output current, for example the Amass XT30 or Molex Mini-Fit connectors.

- A heatsink must be installed for power outputs above 40 W (estimated). The screw hole arrangement is specifically designed to work with 1/8th brick DC/DC converter heatsinks like the Aavid 241802B92200G or Wakefield-Vette 567-45AB. **An insulating thermal pad must be used between the heatsink and the PCB to prevent conduction through the soldermask.** The recommended thermal pad for this module is the Berquist Gap Pad 1500 at 1 mm or 0.5 mm thickness, cut into a 20 mm × 40 mm rectangle.

- The module should be mounted to the vehicle or system chassis with four M3 screws and four M3 nylon standoffs. Use metallic fasteners at your own risk, as they conduct from the exposed power input and power output solder pads.

- To mitigate the risk of conduction from the power input pads on the edges of the module, you may surround the input-side of the module with kapton tape. Only use electrical tape as a protective layer on top of kapton tape for environmental resistance; electrical tape alone weakens with heat and will not completely prevent conduction to exterior materials.

- It is not recommended to surround the entire module and heatshrink with an enclosure, as the design is intended to operate with natural convection cooling. Doing so will trap heat within the assembly and elevate the MOSFET junction temperatures above their maximum ratings.

- Wires introduce additional parasitics and can hamper the supplied voltage to the load. Parasitic resistance introduces a voltage drop across the wire, and becomes worse with bigger loads. Additionally, parasitic inductance in the wire creates much larger voltage drops during transient load events. To keep parasitics to a minimum, use the appropriate wire gauge for your intended maximum load, and keep the VOUT- and VOUT+ wires close together throughout their run to the load.

- The module is susceptible to damage when under vibration. Specifically, the large 1210 ceramic capacitors in the design may fracture with vibration, and may lead to a short-circuit. Take the necessary precautions to reduce the vibration experienced by the module, and **always inspect for cracks in the capacitors before use**.

- While the design features mitigation strategies to improve module resilience against inrush current, these strategies do nothing to actually prevent inrush current. Unless you want to see sparks whenever a battery is plugged in, it is recommended to have a separate system to address inrush current, such as pre-charging through an XT-90S anti-spark connector, or through source-side soft-start circuitry.

# 3 Motivation

In the past, the UBCO Aerospace UAV Division had used 6S LiPo battery packs for powering competition vehicles. The 25 V nominal output voltage allowed our electrical subteam to use the widely accessible LM2596 step down module as "battery-eliminator circuits" (BECs) instead of using a secondary battery to power auxiliary electronics. The modules' 42 V max input voltage, 3 A adjustable-voltage output, and *very* cheap availability eased the team's development process and allowed us to spend less time working on power distribution and more time on addressing competition designs.

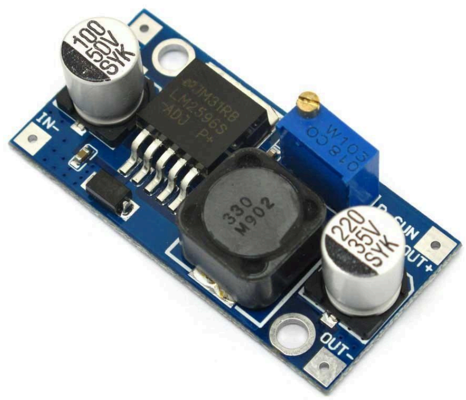


Figure 3.1: The trusty LM2596 module.

Since 2024, the UAV Division has migrated 12S (50 V) Li-ion battery systems for powering their competition vehicles. The higher voltage compared to previous 6S LiPo systems allows for the use of heavy-duty motors, improving vehicle thrust-to-weight ratios and payload capacity. This system comes with the tradeoff of requiring a brand new power architecture, since the 50 V operating voltage exceeds the LM2596's ratings. Additionally, trends in the latest competitions have favoured partially or fully automated UAVs, and has led to the need for more powerful on-board computers that require more than the 3 A typically provided by LM2596 modules. These changes in the overall UAV system architecture are what motivate the design of a new step down module for UBCO Aerospace.

## 3.1 Prior Solutions

Before the UAV division began using 12S battery packs for its drones, the club had already been using the HW-636 and HW-636B, a 60 V-rated, hobbyist step-down converter allegedly capable of providing up to 10 A output current. The primary motives for choosing the HW-636 were its small form-factor and low cost compared to other options. The electrical subteam encountered issues with the HW-636, most notably catastrophic failures when adjusting the output voltage and destructive inrush current during battery connection. Integration tests also revealed poor voltage regulation once the output current exceeded 3 A, causing brownouts with our Raspberry Pi 5 onboard computer.

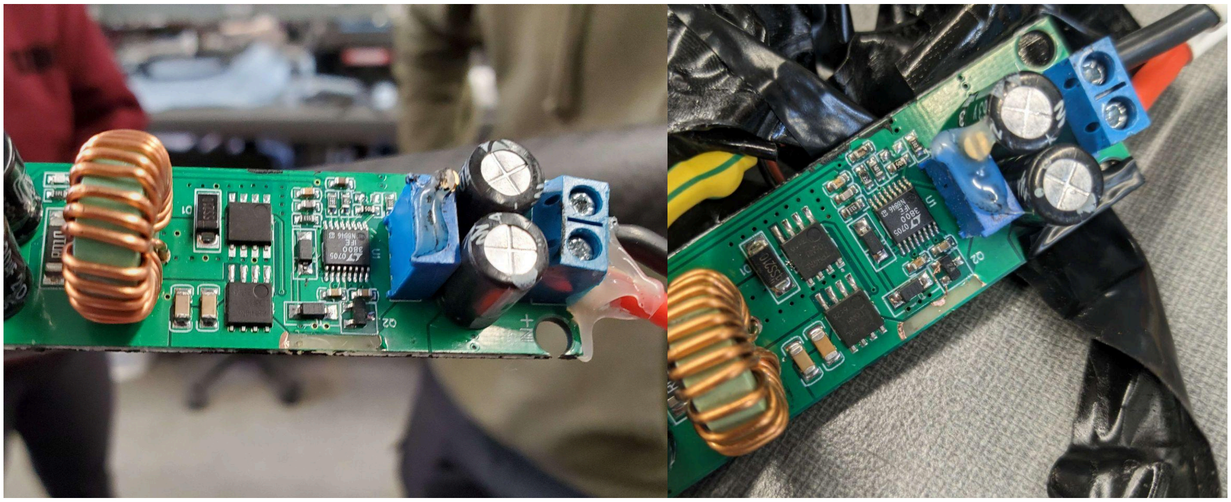


Figure 3.2: Two HW-636 modules after observed failures in 2025. The input trace on both modules exploded immediately after plugging in our 12S batteries.

Reliability issues motivated the team to design a custom solution using the Vishay SiC461 buck converter IC (*which, contrary to the name, does not contain silicon carbide FETs*), dubbed the Super Step Down V2 (or SSDV2). The converter would provide a 5 V bus to power servos and a Raspberry Pi 5. In spite of our efforts, testing revealed severe output ripple and poor regulation, with voltage deviations of up to ±10% and ripple approaching 560 $mV_{PP}$ even under light load. During system integration, the output varied between approximately 4.8 V and 5.3 V, repeatedly causing brownouts at the Raspberry Pi and rendering the design unsuitable for powering our avionics.

Root cause analysis identified insufficient output capacitance as the primary failure mechanism. A misinterpretation of the Vishay PowerCAD reference design led to the omission of twelve out of thirteen specified output capacitors, leaving the converter with much less than the required capacitance and severely degrading transient response and regulation.

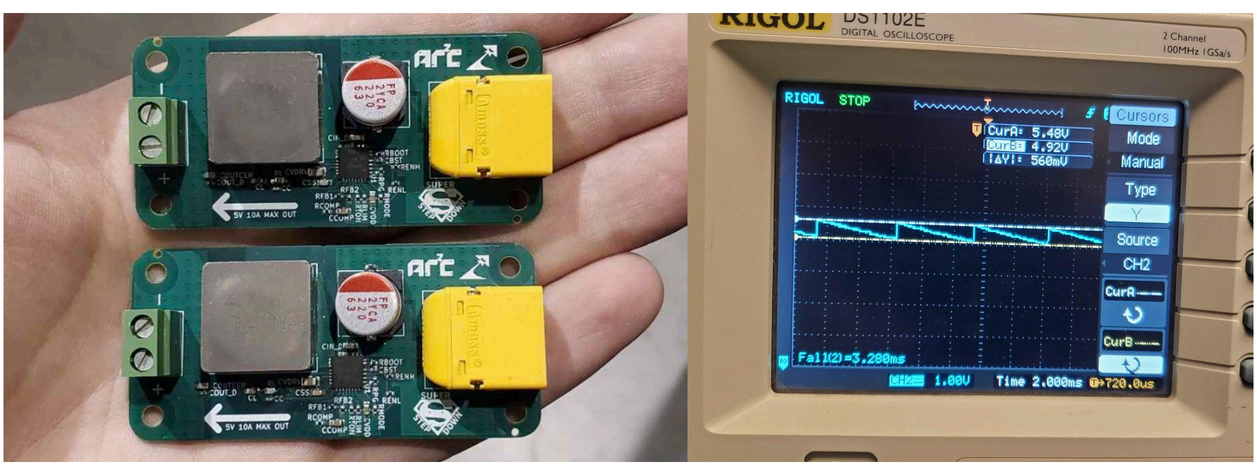



Figure 3.3: The SSDV2 fully assembled. Its design forgot to include output capacitors.

# 4 Circuit Design

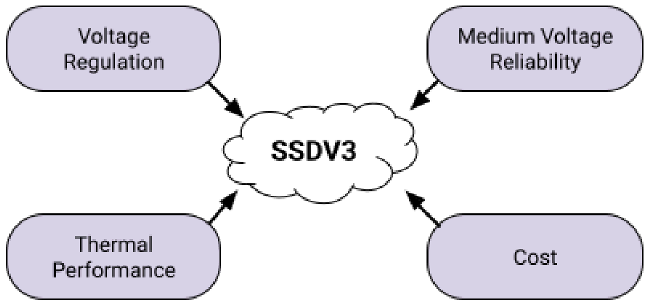



Figure 4.1: Major design factors of the SSDV3.

The main goal of the SSDV3 is to provide a well-regulated voltage under any condition, thus emphasizing the importance of designing for transient performance. Pure no-load measurements have not been an effective metric for testing our step-down modules. Our prior failures with both the HW-636 and the SSDV2 underline transient performance as the most critical design factor of the converter.

This design must also be resilient against inrush currents given the medium operating voltage. Our experience with the failures of the HW-636 show that ceramic capacitors on the input can easily result in inrush currents over 100 A upon sudden connection of a 50 V source, and that the precharge functionality of anti-spark XT90s cannot be relied upon to dampen such currents. As a result, the SSDV3 must be able to handle these inrush currents either through input damping or large copper pours.

While efficiency is not considered the top priority for this design, thermal shutdown or runaway could be a catastrophic failure mode if not accounted for. Our team previously did not make any considerations for thermal performance, other than noting the SSDV2 was very hot to touch during operation. Recent testing at 50 V input showed the MOSFET packages on the HW-636 reached 100 °C within 3 minutes when outputting 12 V at just 3 A, with the junction temperatures estimated to be operating close to 115 °C. The converter must be cooled solely through natural convection since our aircraft are usually designed with no chassis airflow.

Cost must be considered through the design, since funding across UBCO Aerospace has been spread thin for 2025-26. The component selection and board requirements must be carefully balanced with the overall unit cost in order for the production of the SSDV3 to be attractive over other commercial off-the-shelf solutions (COTS). This means restricting PCB specs to 4-layer, 1 oz copper, and taking advantage of “economic” processes from board manufacturers.

Finally, the solution footprint is a moderate factor influencing the SSDV3 design. Just as with cost, the size of the design must be competitive to other COTSs and the prior SSDV2.

## 4.1 Requirements

| Parameter | Value | Description |
|---|---|---|
| $V_{IN}$ | 20-70 V | Allow for either 6S or 12S battery input, plus ample margin |
| $V_{OUT}$ | 5.1 V, 7.4 V, 12 V | Selectable output for the common voltages |
| $I_{OUT,MAX}$ | 8 A | Abs. worst case current draw from Raspberry Pi 5 plus margin |
| $f_{SW}$ | 300-500 kHz | Design for transient performance and small footprint |
| $\Delta V_{OUT}$ | <5% | Maximum output voltage deviation during transients |
| $\Delta T_{J,MAX}$ | +80 °C | Maximum junction temperature rise |
| L | <65 mm | Length is smaller than the SSDV2 |
| W | <32 mm | Width is smaller than the SSDV2 |

Table 4.1: Quantified design requirements.

## 4.2 Converter Topology

The most suitable and simplest DC/DC converter topology for this design is the buck converter. More sophisticated topologies like the flyback or forward converter would not offer any benefit at the required power output, and the use of a transformer would only complicate the design. The solution space for buck converters at this power output can be broken down into three major categories:

- a buck converter IC with integrated FETs,
- a buck controller with external FETs, and
- a dual-phase buck converter

A buck converter IC represents the simplest option of the three, removing the challenge of choosing discrete MOSFETs and allowing for the smallest footprint. While a number of these buck converters are rated for 8-10 A output, their efficiency curves suggest total system power losses of 3-5 W, very likely requiring heatsinks in order to operate at these currents for prolonged periods. Granted, the losses from parasitics in an integrated buck converter will be lower than a converter with external FETs [1], but attempting to improve such high thermal dissipation in a single package will be challenging. The most promising IC at this time is the LM70880 buck converter from Texas Instruments (TI), optimized for high step-down-ratios, though also the most expensive option at $11.10 CAD per unit from DigiKey. Other solutions like the MIC28516, LM76005, or SiC461 either are too inefficient at rated current, offer little voltage margin at
$V_{IN}$ = 50 V, or use a variable-frequency control scheme that is not easy to design around.

Designing a buck converter using a controller and external FETs provides a greater amount of flexibility in the design, particularly in optimizing for either switching or conduction losses. Using discrete FETs also offers the benefit of greater heat spreading across the PCB, but comes with the cost of a more laborious PCB layout and necessitating more design time for selecting FETs. The selection of buck controllers is only limited by the 70 V max input voltage and minimum on-time specification of at least $t_{ON} \approx$ 5 V / 70 V / 500 kHz = 140 ns [2]. Once again from TI, the LM5146 and LM5148 buck controllers appear to be the most attractive options, at $2.49 and $2.25 CAD on LCSC respectively. With most 100 V MOSFETs costing about $1.50 CAD each, we could potentially design a discrete-FET synchronous buck converter for nearly half the cost of the LM70880! The cost alone makes a discrete-FET buck converter the most attractive approach.

Using a dual-phase buck converter reduces both the ripple voltage and transient deviation of a typical buck converter. In fact, modern motherboard voltage regulator modules (VRMs) are implemented as multi-phase buck converters that are specifically optimized for fast transient CPU loads, suggesting that a similar approach is well-suited for supplying embedded computers like the Raspberry Pi 5. Two phases also hypothetically reduces the solution size, since two small inductors are typically smaller than one large inductor [3]. The compromise of this approach is it requires twice as many FETs compared to a single phase converter, and possibly two controllers as well. A number of dual-phase buck reference designs exist from TI for both the LM5148 and LM5143 controllers. While these designs show that a multi-phase buck converter can reduce the number of passive components, the overall solution size is typically larger than that of a single-phase buck converter and provides substantially more power than required for this application.

Evidently a buck controller with external FETs was chosen for the topology of this design.

## 4.3 PWM Control Method

At a glance, both the LM5146 [4] and LM5148 [5] equally meet the design requirements set out in 4.1. Choosing between the two ICs comes down to deciding whether the buck converter PWM generation should use voltage-mode control (VMC) with line feed-forward or current-mode control (CMC). This decision must be made early in the design process to determine the desired inductor current ripple factor.

An abhorrent amount of my time has been spent researching the most optimal control mode, with what felt like hundreds of research papers, conference papers, and application notes. The most useful resources were [6-13] for making this decision. In general:

- Current-mode control simplifies the plant by changing the output stage LC double pole to a single RC pole. This is desirable for topologies with right-half-plane-zeros (not the buck converter.)
- Both VMC and CMC can provide good line rejection. In spite of praise for CMC's "intrinsic feed-forward" sensing via inductor current for line variations, VMC with feed-forward is still encouraged for applications with large line transients without much justification.
- Voltage-mode requires more compensation circuitry compared to current mode.
- Current-mode control requires current sensing which can be susceptible to interference. Sensing may also require blanking time to eliminate leading-edge current spike, increasing high-side switch minimum on-time (though there is no mention of blanking time in the LM5148 datasheet).
- Current-mode control may suffer from subharmonic oscillations for duty cycles above 50% (which isn't very applicable to this design.)
- The LM5148 requires the inductor to be specified to match the appropriate inductor current ripple downslope for its internal slope compensation, with the ideal inductor calculated using Equation 9 from the datasheet [9]. This specification can conflict with the rule-of-thumb inductor current ripple factor of 0.2-0.4.

*The tradeoffs between the two control techniques can be discussed endlessly and is not the subject of this section.*

The deciding factor for choosing the controller was the ease of understanding the small-signal transfer function of each control technique. The described voltage-mode analysis in [14-16] was easier for me to understand and implement compared to the current-mode analysis in [17] and [18]; additional design parameters for slope compensation and modulator gain made derivation of the control-to-output transfer function unclear to me at the time. Hence the LM5146 was selected.

## 4.4 Power Stage

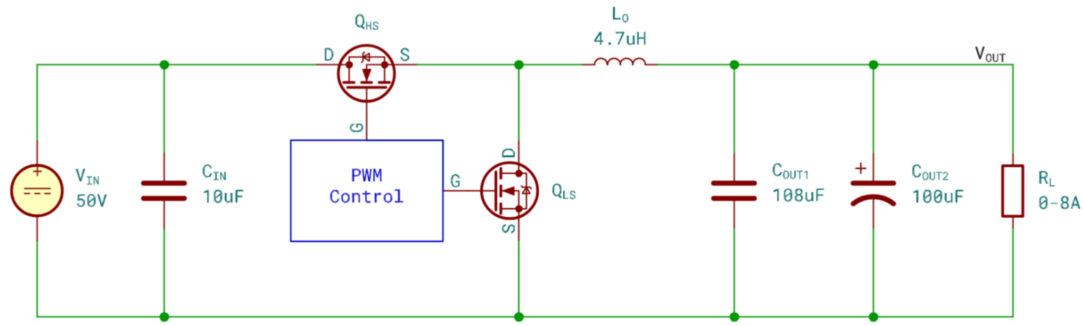



Figure 4.2: The SSDV3 simplified power stage.

The power stage of the buck converter consists of the input capacitors, the high-side and low-side switching MOSFETs, the inductor, and the output capacitors. The component values were selected sequentially, beginning with switching frequency and desired inductor ripple factor, then the MOSFETs, followed by output capacitance sizing based on output ripple and transient constraints, and finishing with input capacitance selection for input ripple and bulk energy considerations.

The calculations for the power stage components used equations from [19] and [20], alongside assistance from the LM5146/-Q1 quickstart design tool available from TI.

### 4.4.1 Switching Frequency and Inductor Selection

One of the critical design parameters of a buck converter is the inductor current ripple, which is usually expressed as a ratio to the maximum output current, and it can only be controlled through the choice of inductor and switching frequency. A large current ripple causes higher power loss in the inductor through hysteresis loss and increased output voltage ripple, while a ripple that is too small requires a very big inductor and hampers voltage regulation due to limited current slew rate (illustrated in Figure 9-1 of the LM5146 datasheet [4]). A current ripple of 30% is widely used as a rule-of-thumb for buck converters and is set out as the desired ripple for this design.

Inductor current ripple is strongly dependent on the converter duty ratio, which will vary for the different output voltages of 5.1 V, 7.4 V, and 12 V. Since it will not be possible to optimize the current ripple for all three output voltages with voltage-mode control, a compromise is made, and a 30% current ripple is only designed for the 5.1 V output voltage since it is expected to power the most sensitive equipment. This compromise results in the 12 V output mode having a ripple current of about 50%, which has the benefit of not hindering the transient performance of the converter at higher voltages, at the cost of power loss and greater output voltage ripple.

Since this design is intended to be assembled abroad, the choice of components must be optimized for availability with Eastern distributors like LCSC. A quick search for parts from well-known manufacturers on LCSC show that only E3 series inductors are usually well stocked. Knowing that the required switching frequency range is limited by the effects of switching power loss, and that larger inductors are also bulklier and carry higher resistance, a 4.7 µH inductor is chosen as the optimal balance between all factors. Consequently, the switching frequency is specified to 420 kHz to achieve a 30% current ripple at $V_{OUT}$ = 5.1 V.

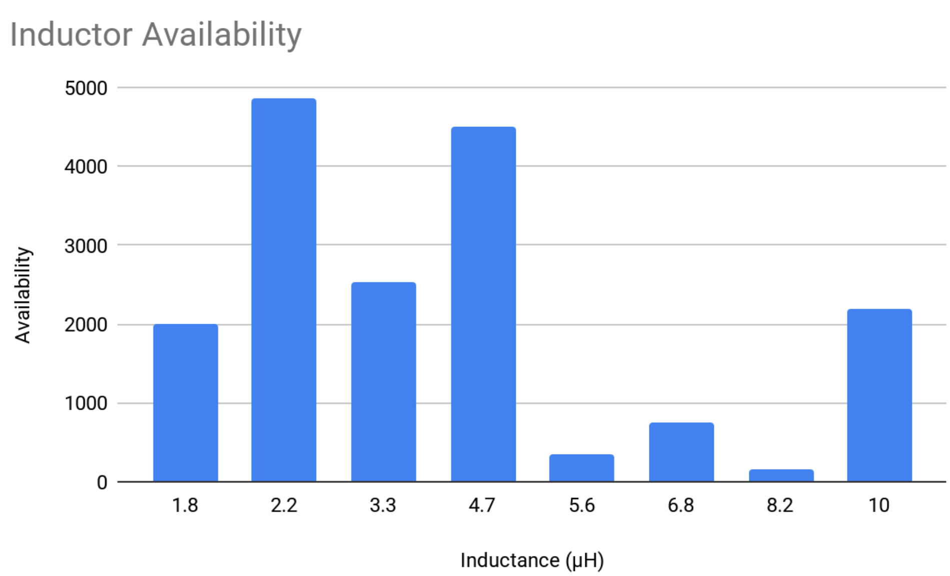



Figure 4.3: The availability of different inductor sizes on LCSC that meet the design specifications.

### 4.4.2 MOSFET Selection

Typical textbook literature about MOSFETs in switching applications describe two sources of power loss: conduction when the MOSFET is on, and losses in turning the MOSFET on or off. Guidance from [21], [22], and the equations 10-12 in [23] reveal a third source; $Q_{oss} + Q_{rr}$ transition losses between the low-side MOSFET parasitics (also called the synchronous rectifier) and high-side MOSFET turn-on event can easily dominate over the other losses at the intended 50 V input voltage, since they are directly proportional to the input voltage:

$$P_{\text{HS turn-on}} \approx (Q_{oss} + Q_{rr})_{LS} \times \mathbf{V_{IN}} \times f_{SW}.$$

Further complicating things, both [22] and [24] explain that the $Q_{oss}$ and $Q_{rr}$ datasheet specifications are strongly dependent on the exact MOSFET drivers, and that the test conditions in the datasheet seldom align with real world switching applications, even if the test voltage matches the intended operating voltage. Unfortunately UBCO Aerospace does not have the necessary test equipment to independently verify the $Q_{oss}$ and $Q_{rr}$ parameters of each MOSFET candidate, so the datasheet specifications will have to be taken at face value for comparisons.

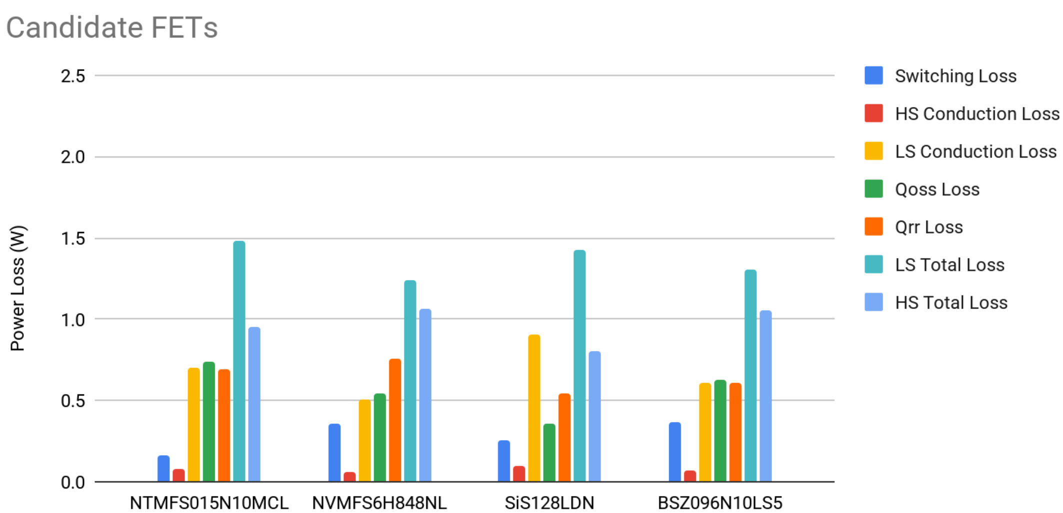



Figure 4.4: Power loss breakdown of some candidate MOSFETs.

The figure above analyzes the power losses of some of the best candidate parts that were available on LCSC for Revision C, using the methodology described in [23]. The BSZ096N10LS5 [25] from Infineon was chosen as the low-side MOSFET for its high availability and equally distributed losses across each category compared to the other options. However, the relatively high singular unit cost of $1.93 CAD through LCSC motivated choosing a different part for the high-side MOSFET. With a simpler ($R_{DS(on)} \times Q_g$) figure-of-merit, the CSD19537Q3 [26] from TI was chosen for the high-side MOSFET with a very competitive $1.12 CAD singular unit cost.

Significant $BV_{DSS}$ derating of 60-100% from the nominal operating voltage of 50 V was chosen to account for regenerative back EMF produced by UAV motors during flight [27].

### 4.4.3 Capacitor Selection

The selection of output capacitors is one of the more nuanced elements of the design process for this project. The output capacitors are the defining factor for output ripple and transient performance, and also impact the later controller implementation through their contribution of a pole-zero pair to the power stage transfer function [28]. Generally more capacitance with a low ESR improves all quantitative factors of a buck converter in exchange for greater production costs and board space.

This design takes a hybrid approach and uses a combination of ceramic and polymer tantalum capacitors to simultaneously address both voltage ripple and transient performance, achieving both low ESR and high energy density in a small form-factor. This approach takes inspiration from the consumer graphics cards, where power filtering uses a combination of ceramic and tantalum capacitors to meet the strict voltage requirements of GPU chips [29]. The drawback of using two different capacitor chemistries is it complicates the power stage transfer function by adding another pole and zero to the system [30]. To keep the design simple, the value of the ceramic capacitors is roughly matched to the size of the tantalum polymer capacitor, resulting in the added pole and zero to roughly cancel out.

The output capacitance is realized using one KEMET T521D107M025ATE040 100 μF KO-CAP tantalum capacitor, and six Samsung CL32A226KAJNNNE 22 μF X5R ceramic capacitors, with an effective capacitance of 108 μF at 5 V and 53 μF at 12 V. A polymer tantalum capacitor was chosen over a polymer electrolytic capacitor to keep the module size compact.

The input capacitors are important for minimizing input ripple and switch-node parasitics. In this application only ceramic capacitors are used at the input for their low ESL and small footprint, minimizing the design's parasitic loop inductance. Four Murata GRM32EC72A106KE05 10 μF, 100 V rated ceramic capacitors implement the input capacitance, with total effective capacitance of 10 μF at 50 V. An additional 100 nF 0603 capacitor is used to keep the switching current loop tight near the power stage MOSFETs and minimize loop inductance.

An additional 47 μF electrolytic capacitor is added to the input only for the purpose of damping wire resonances.

## 4.5 Controller Implementation

### 4.5.1 IC Power Input

The power input to the LM5146 buck controller uses a 2.2 Ω series resistor with a 100 nF decoupling capacitor to provide minor high frequency filtering from the 50 V input line.

### 4.5.2 UVLO Implementation

Accounting for converter dropout voltage, the UVLO is set to 15 V, or about 3 V above the maximum output voltage of 12 V.

### 4.5.3 Soft Start

The anticipated operating conditions of the converter does not involve highly capacitive loads, so a modest 4 ms soft-start period is implemented through a 47 nF capacitor on the SS pin.

### 4.5.4 Bootstrap Circuit

The high-side gate driver uses a bootstrap circuit to keep the high-side MOSFET on by allowing the gate voltage to momentarily swing above $V_{IN}$ for the duration of the switching on-time [31].

The chosen high-side MOSFET does not present a demanding switching action under regular operation to the LM5146's internal gate driver circuitry, so a typical 100 nF capacitor is sufficient. A 2.2 Ω series resistor is also used to slow down the turn-on time of the high-side MOSFET and consequently reduce switch-node ringing [32].

### 4.5.5 OCP Implementation

The OCP threshold was arbitrarily defined as roughly 50% of the maximum drain current of both MOSFETs at 125 °C, specified by their datasheets. The threshold is hence constrained by the CSD19537Q3 with a max $I_D \approx 23$ A at 125 °C [26], so the OCP target threshold is set to 11.5 A.

The OCP is implemented using the low-side MOSFET $R_{ds(on)}$ sensing method defined in the LM5146 datasheet. The $R_{ILIM}$ resistor is set to 430 Ω, an E24 series value, alongside a 12 pF C0G decoupling capacitor to address the 6 ns RC time constant requirement from the datasheet.

### 4.5.6 FPWM/DEM Mode

With the objective of excellent transient performance under all conditions, FPWM mode is set for light-loads. This is implemented using a 20 kΩ pull-up resistor on the SYNCIN pin to the IC VCC rail. In the event that DEM mode is desirable, a 0 Ω resistor can be placed on the DNP'ed R8 to pull the SYNCIN pin to GND.

## 4.6 Feedback and Loop Compensation

Voltage-mode buck converters require a type-III compensator to achieve closed-loop stability [33], shown in Figure 4.5. An integrator pole and a pole-zero pair are introduced through the compensator feedback impedance on the op amp, internal to the LM5146. A second pole-zero pair is added through the input impedance to the op amp inverting input.

Compensation is conventionally achieved through loop shaping–placing the compensator zeroes near the power stage resonance frequency, a compensator pole near the power stage output capacitor ESR zero, and another pole at $f_{SW}$ / 2 to achieve the desired gain and phase margin [14].

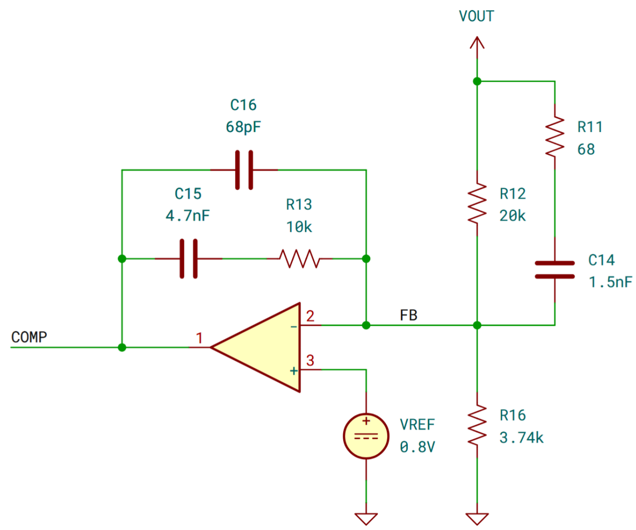



Figure 4.5: The type-III compensator circuit.

The TI-provided LM5146/-Q1 quickstart design tool includes fields for automatically designing the compensator, but the tool is only intended for a single type of output capacitor. Hypothetically this should be fine for this mixed output capacitor design, since the different output capacitors were specifically selected so the introduced poles and zeroes would remain simple. But DC bias effects on the ceramic capacitors meant the power stage output double-pole would migrate in frequency with different output voltages (e.g. 7.4 V and 12 V).

The compensation-tuning method had become somewhat crude. To preserve the convenience of the quickstart design tool, the control-to-output transfer function was first derived in Figure 4.6 and used in MATLAB for the various output capacitors. An equivalent single-capacitor model was then estimated by inspecting the bode plot and pole-zero locations, and the resulting ESR and capacitance were entered back into the tool. The tool's compensator parameters were finally re-checked in MATLAB to verify open-loop behavior and stability margins. This process produced a compensator with robust stability across all output voltages.

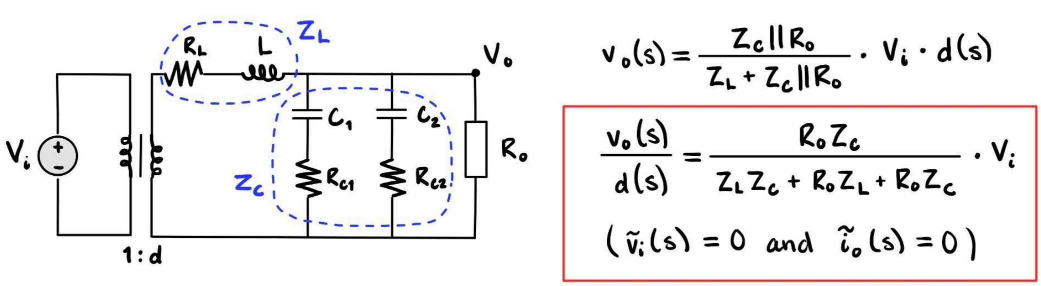



Figure 4.6: Derivation of the control-to-output transfer function using the state-space averaged buck converter model and generalized capacitors. Input voltage and output current perturbations are zeroed.

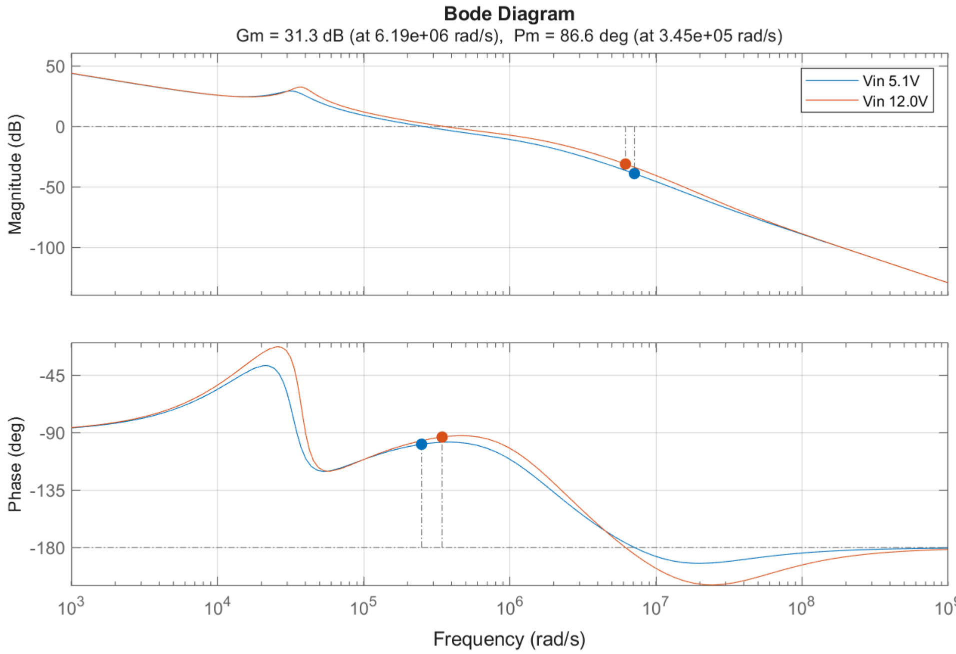



Figure 4.7: The full open-loop transfer function bode plot, after compensation.

Figure 4.7 illustrates the bode-plot and stability margins of the open-loop system with compensation. The 5.1 V output has a phase margin of 80.9° at a crossover frequency of 39.6 kHz. The 12 V output has a phase margin of 86.6° at 54.9 kHz. While the stability margins seem extreme, well above the 60° phase margin typical for robust control, the migrating double-output pole caused by the DC bias effect on the output capacitors made it difficult to balance the phase with the target crossover frequency of $f_{SW}$ / 10 through compensation. The aggressive margins are simply the result of attempting to balance the control bandwidth with loop stability.

Though unmentioned in the motivation section of the documentation, UBCO Aerospace encountered one noted issue with the SSDV2 where the output voltage seemed to drift over time as the module provided power to a demanding load. It is hypothesized the voltage drift was caused by the changing of value on the thick-film feedback resistors from temperature drift as the SSDV2 heated up [34]. To prevent this issue from surfacing again in the SSDV3, the lower feedback resistors on the feedback voltage use thin-film resistors with very low temperature coefficients of 50 ppm/°C or lower. This is why the voltage-setting resistors use E96 series resistors, while every other resistor is from the E24 series.

On a smaller note, the user sets the desired output voltage by shorting jumper pads that “enable” the lower resistors in the voltage divider. If each target voltage of 5.1 V, 7.4 V, and 12 V had its own dedicated lower resistor, a user could accidentally short both the 7.4 V and 12 V jumpers, producing a much higher output voltage than intended. To prevent this, the lower resistors are chosen so that achieving 12 V requires shorting both jumpers. The different output voltage configurations based on the shorted jumpers is shown on the silkscreen of Figure 2.2.

## 4.7 Reverse Polarity Protection

The reverse polarity protection uses an example circuit shown in Section 8.2.6 of the LM5050-1 datasheet [35], and is illustrated in Figure 4.8. While the LM5050-1 is intended as an ideal diode controller, its purpose in this circuit is to drive the N-channel MOSFET during regular operation.

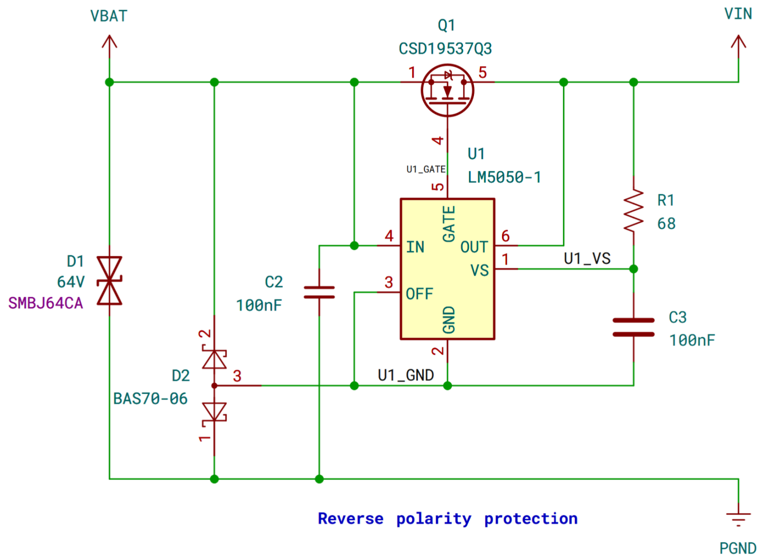



Figure 4.8: The reverse polarity protection circuit, implemented with the LM5050-1. Though not included, the internal body diodes of the power stage MOSFETs are a part of this circuit.

The working principle of the circuit is that the internal body diode of the Q1 MOSFET allows the LM5050-1 and downstream buck converter to power on, then the LM5050-1 turns on Q1 to allow for conduction through the FET N-channel region rather than the body diode.

In the event of reverse polarity on VBAT, 50 V is now connected to the PGND net. The diode network D2 prevents backwards conduction through U1, and the body diode of Q1 blocks significant backwards conduction through the downstream buck converter. **Some necessary conduction still occurs to the VIN net via the internal body diodes of the power stage MOSFETs**, which replaces the D2 SMBJ60A diode shown in Figure 31 of [35] operating in forward conduction during reverse polarity.

Since the VIN net is also at 50 V, the LM5050-1 powers on normally, and the U1_GND net's return path is redirected by the D2 diode network back to the battery through the VBAT net (which is now ground). With the OUT pin having a higher voltage than the IN pin, the LM5050-1 keeps Q1 turned off, fully preventing reverse conduction and ensuring the whole system remains in a well defined state when in reverse polarity.

*Why are we using bidirectional TVS diodes* before *the RPP circuit in contrast to Figure 31 in the LM5050-1 datasheet? Because the bidirectional TVS diodes were cheaper than the unidirectional ones on LCSC! And, as stated, the power stage MOSFET body diodes already serve in that role.*

# 5 PCB Design

## 5.1 Board Specifications

A 4-layer, 1 oz/ft$^2$ copper thickness PCB is used for this design. The typical differential stackup of common 4-layer boards helps reduce the ground return path distance and offers better EMI mitigation compared to a 2-layer board. An aggressive 0.25 mm trace clearance is set to improve manufacturing tolerances, and a further 0.5 mm clearance was used on 50 V exterior traces.

The solder joint integrity of the QFN packages used in this design are sensitive to the uneven pads produced by HASL surface finishes [36]. So instead an ENIG surface finish is used.

## 5.2 Grounding

The entire inner layer L1 is used as a unified ground plane, while all other layers use ground-fill to improve the thermal dissipation of the board. Stitching vias are used close to the power stage capacitors to reduce their return path impedance.

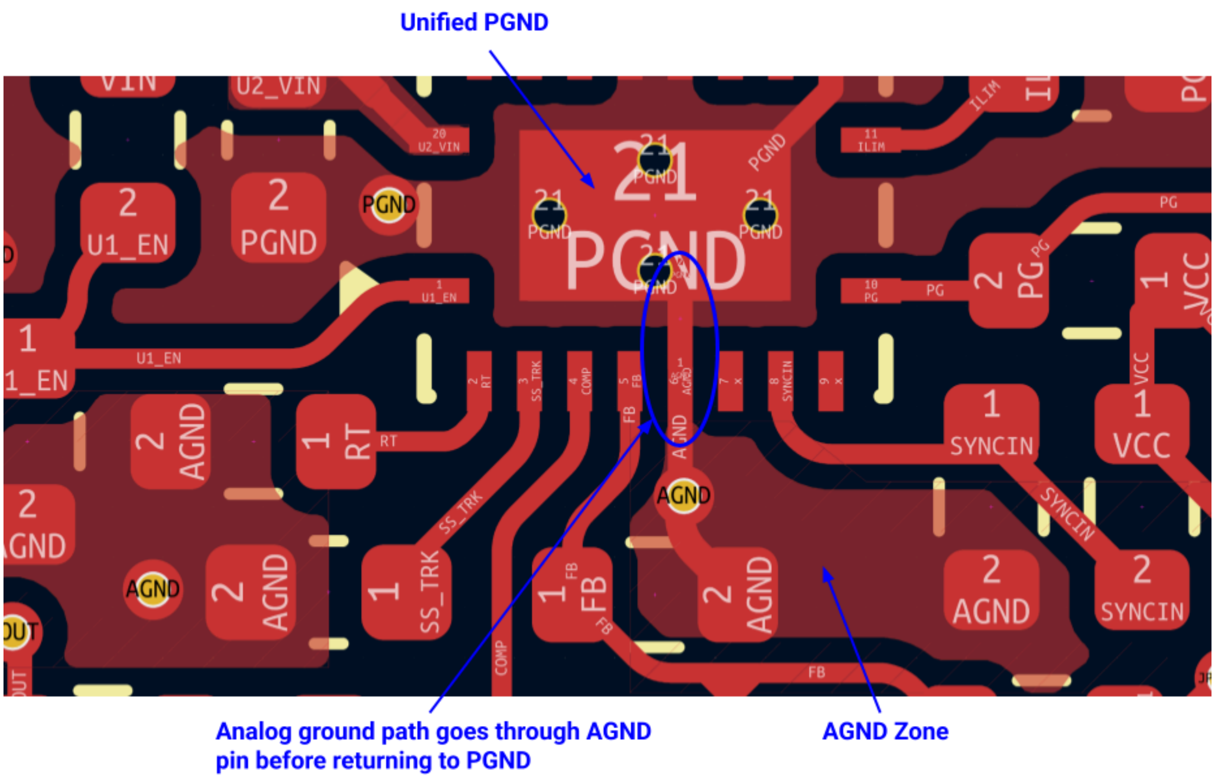



Figure 5.1: The PGND and AGND routing.

The analog ground net is connected via net-tie to the unified ground plane in close proximity to the LM5146 AGND pin. The purpose of the separate analog and power ground nets is to control the return path of the analog circuits so that they flow through the AGND pin while still remaining connected to the unified power ground plane, as shown in Figure 5.1. They are not separated for the purpose of complete circuit separation [37].

## 5.3 High-frequency Power Loop

The LM5146 datasheet (in Figure 11-1 of [4]) and numerous application notes emphasize the importance of proper layout for the power stage of buck converters, mainly as the primary method for minimizing the system's conducted and radiated EMI. While the anticipated EMI from this buck converter is not a major concern in drone applications, it is still important to manage the power stage layout to control the switch-node ringing and peaking caused by parasitics. Furthermore, the fast *dv/dt* switching can also induce "phantom turn-on," where the low-side MOSFET gate voltage is momentarily forced high through drain-gate coupling of the MOSFET miller capacitance, causing it to turn on and result in high-current shoot-through events [38]. The implemented layout is shown in Figure 5.2.

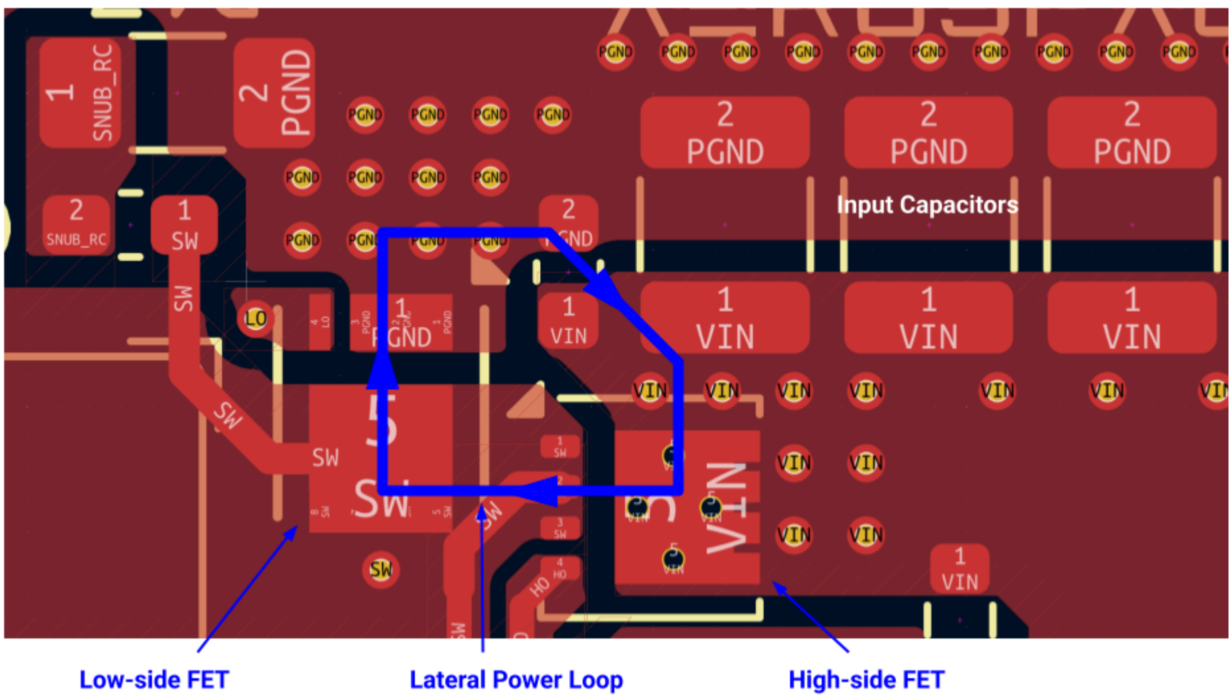



Figure 5.2: The MOSFET placement and power loop.

The SSDV3 Rev. C layout uses the "improved lateral switching loop design" discussed in [39], where the MOSFETs are placed perpendicular to each other. The particular selected MOSFETs use 3.3 × 3.3 mm PQFN packages that are smaller than the popular 5 × 6 mm SON "SuperSO-8" packages, allowing for a smaller power stage. Combined with a 0603 ceramic capacitor between the two MOSFETs for high-frequency decoupling, the resulting enclosed area of the lateral switching power loop is kept to a minimum.

If the power stage layout design fails, and the switch-node ringing is still unsatisfactory, a basic RC snubber circuit has been included in the design, whose components are to be added experimentally in testing to reduce ringing. A 0 Ω 1206 resistor is used to connect the snubber copper island to ground, preventing the unused snubber copper from floating.

In a similar fashion, the gate driver circuitry from the LM5146 to the MOSFETs is sensitive to the PCB design. Because of the LM5146's ability to source and sink 2-3 A to and from the MOSFET gate in the span of nanoseconds, parasitic inductance in the gate driver path can cause slower turn-on times and unwanted ringing. Usual advice to mitigate the latter is to use gate resistors to dampen any ringing, yet carries the downside of further slowing the MOSFET turn-on times and introduces losses. The SSDV3 Rev. C already features a resistor in the bootstrap circuit to slow the turn-on of the high-side MOSFET, so adding gate resistors would be redundant and counterproductive. Instead, the PCB design focuses on minimizing the parasitic inductance in the gate driver path using the layout shown in Figure 5.3.

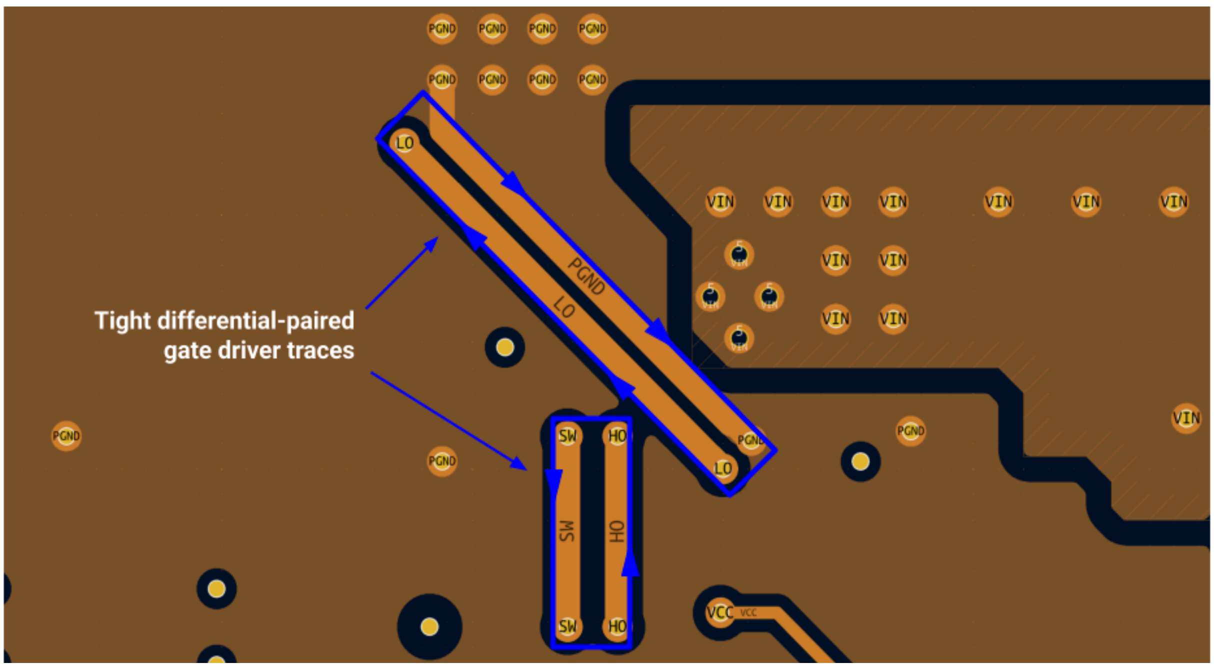



Figure 5.3: The gate driver trace routing on the internal layer L2.

The high- and low-side gate drivers are routed differentially with their return paths to minimize loop inductance. Wide 0.5 mm traces ensure the resistance from the gate driver to the MOSFETs is kept low, maximizing the gate drive strength from the LM5146.

## 5.4 Inrush Current Mitigation

The potential damage caused by inrush current is mitigated primarily through extra large copper pours leading to and from the buck converter input capacitors. The width of the copper leading from the battery input pads are at least in excess of 4.00 mm, providing a large copper cross-section for conduction through the whole return path for the inrush current.

Review of the HW-636 copper trace explosions also suggested that resonance in the wires leading to the module increased the net energy dissipation in sensitive areas and contributed to its failure. A 47 μF electrolytic capacitor is added to the input as a means to dampen these resonances.

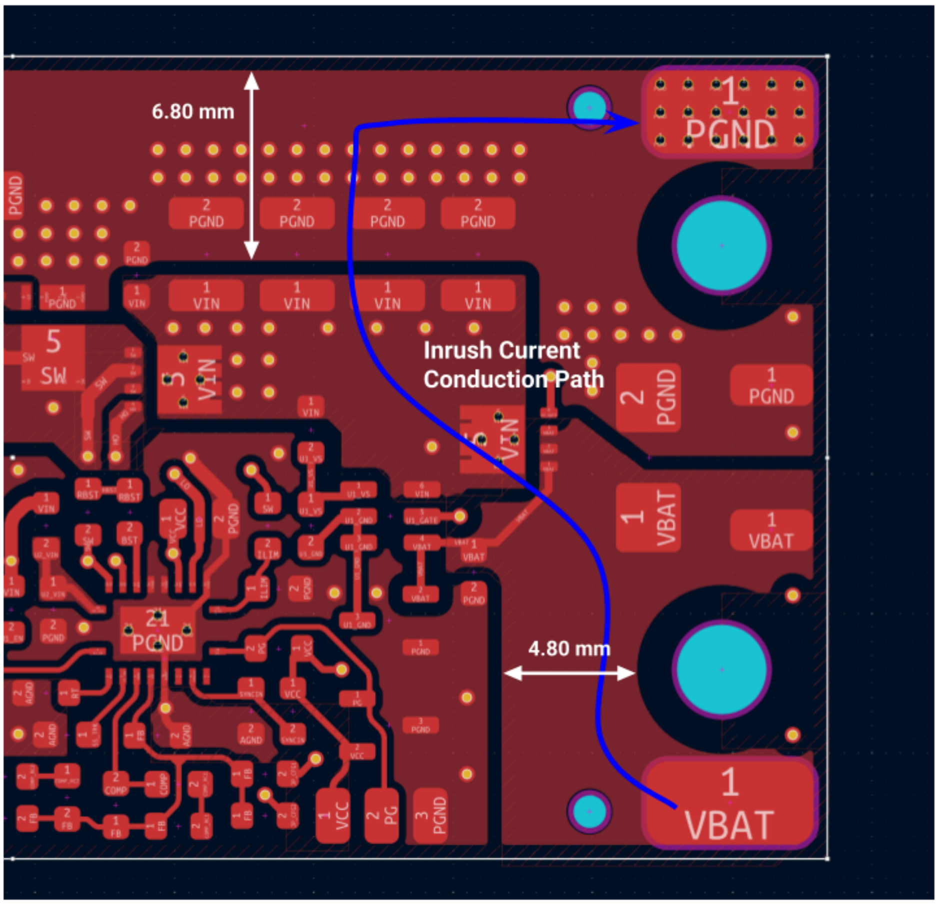



Figure 5.4: The reinforced copper pours to mitigate inrush current.

# 6 Testing Notes

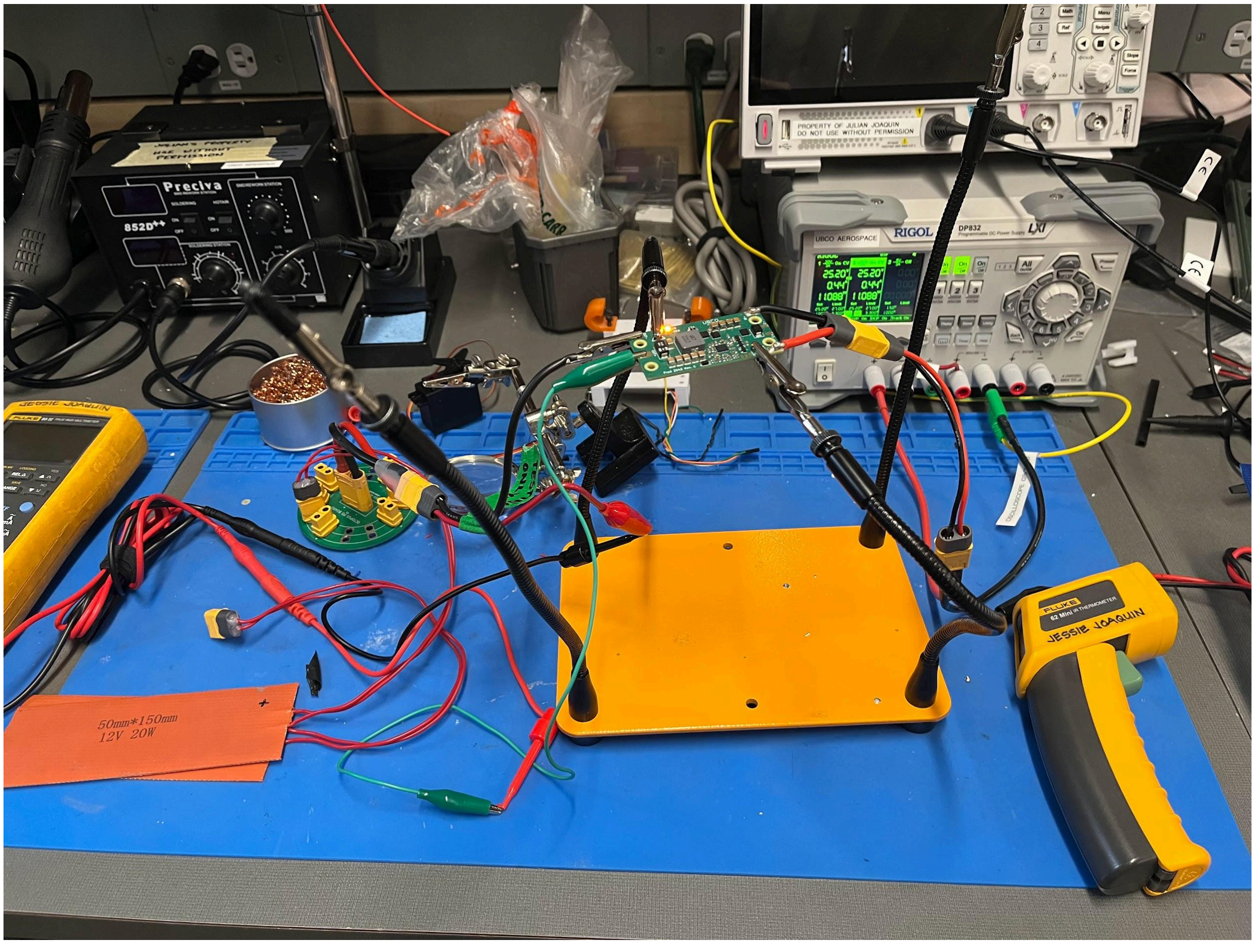



FIgure 6.1: The test rig used for measuring regulation under load and thermal characteristics. Messy, eh?

- UBCO Aerospace does not have access to a professional load tester, so we had to get creative when it came to performing load tests. For DC loads, we had seven heating pads on hand, initially intended for a nitrous oxide tank thermal blanket, and decided to use those as incremental loads. Each heating pad was intended for 20 W at 12 V, or in other words had a resistance of 7.2 Ω. Though this number seemed to vary wildly in testing and we often saw the resistance be in the higher ranges of 8.0 Ω. Put all seven in parallel, and the smallest load resistance we could muster is about 1.15 Ω. This was the limiting factor for all our load testing, and why the 5.1 V and 7.4 V performance curves in Section 1.3 don't extend all the way to 8 A.

- For transient load testing, we created a fairly simple, makeshift circuit using a 555 timer, one typical IRFZ44N MOSFET, and some cheap low-value resistors. The system-level circuit is shown in Figure 6.2. A simple monostable multivibrator drives a MOSFET and provides a conduction path through a 1 Ω, 20 W-rated resistor. A 0.1 Ω low-side resistor is used for shunt-based current sensing via an oscilloscope.

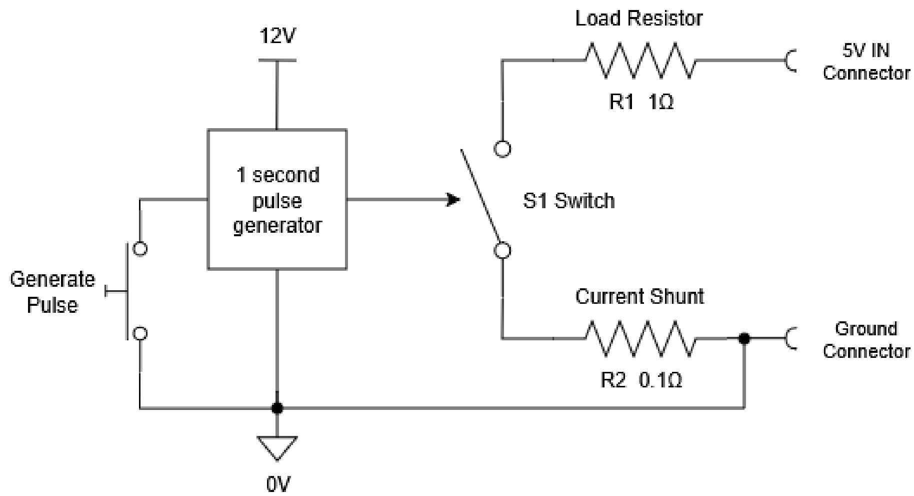



Figure 6.2: The makeshift transient load tester.

- DC load current measurement was conducted using a Fluke 87 IV multimeter with some very shoddy alligator clip test wires. Input current measurements were simply read from the display of UBCO Aerospace's RIGOL DP832 test bench power supply with a limited resolution of 0.01 A.

- Temperature measurements were taken from a Fluke 62 Mini IR thermometer. In order to precisely measure the temperature of the MOSFET packages, the thermometer had to be held mere millimeters above the module under test. At this distance, it was difficult to gauge what exactly the thermometer was measuring, especially since it was clear the built-in laser pointer was intended for distances in meters, not millimeters, and was significantly offset from the actual measurement position at such close distances. *We desperately need a thermal camera*.

- The fast switching rise times introduced noticeable coupling into the oscilloscope measurements throughout testing. At some point, we had mounted the oscilloscope probe's 3 inch ground wire some distance away from the switch-node trace when attempting to measure the switching waveforms. The observed waveforms showed major 65 V peaking on the switching rising edge, alongside some very visible ringing in the tens of megahertz. It took a while to realize the waveforms did not reflect the actual switching voltage, and we relocated the probe ground wire to a closer ground node to finally get the waveforms seen in Figures 1.8 and 1.9. *Keep your probe ground wires close to your measurement points!*

# 7 Discussion

The Rev. C design back in September of 2025 was only intended to be a prototype/testbed for further development, while the Rev. D design would be fully geared towards deployment for UBCO Aerospace. When this project started, it was not even guaranteed the design would be produced since we were more focused on looking for more commercial alternatives to the failed modules discussed in Section 3.1. Our goals shifted once it was realized the Rev. C design would be feasible, and it was rushed into service so that testing of the Jellyfish competition aircraft could fully commence by November. Nonetheless there were many lessons learned from the design that will motivate changes in the next revision.

## 7.1 Lessons Learned and Next Revision

- More thermal mitigation is needed. Heat output from the module is the present limiting factor for the maximum power output, and it is causing the design to fall short of its target output current. This will involve using higher temperature-rated components (e.g. X7R capacitors) and reducing the power loss of the current design when under load.

- Eventually, the design should not require a heatsink. While it seems counterproductive to the prior statement, the use of a heatsink has introduced some mounting challenges that restrict the universality of the design for drone applications. The 1/8th brick-style hole arrangement in particular is not friendly to any common FPV drone screw grids, something only we've solved by using 3D-printed mounts. Many colleagues would also like to eventually be able to encapsulate the module in heat-shrink. Maybe?

- Better connectors are needed on the PCB, as solder pads do not provide any isolation. The idea of using solder pads originated from how most drone ESCs and BECs have directly soldered wires to the PCB. But with the sheer wiring complexity of many of our projects at UBCO Aerospace, the lack of a physical barrier to the conductive parts of the connections has posed a serious risk should a wire come loose. We've had to compensate for this by using a variety of mitigation techniques to prevent conduction never done before in this club, as discussed in the Installation Guidelines. While those guidelines are generally good practice, the design should not have to rely on those practices to achieve safe isolation at the current operating voltage. Additionally, the placement of the solder pads involves non-negligible separation in the power wires, introducing unnecessary loop inductance at both the input and output. So something like an XT30PW connector will be welcome.

- The output ripple shown in Figure 1.6 and 1.7 have a significant ESL spike component in their waveforms, likely tied to the distant placement of the output capacitors away from the low-side MOSFET source pin. Future revisions will need to find a way to reduce the loop inductance of the output capacitors.

- Fix the electrolytic input cap so it's protected by RPP!

## 7.2 Bloopers

- The initial design schematics had circuitry for input hot-swap protection, rather than reverse polarity protection, since we had assessed inrush current to be a major risk. However the limited board space and the additional costs of a hot-swap controller led us to scrap the idea. Reverse polarity protection was only considered after rediscovering some discourse on how some SSDV2 failures were *allegedly* attributed to reverse polarity. The decision to include RPP in the final design came after reading some reverse polarity failures in the Open Source DC/DC Converter project by Rootthecause (see Further Reading).

  > The v5 still had proper reverse polarity protection. However, all successors no longer did (only the intrinsic body diodes). Since the space requirement and (depending on the implementation) loss of power could not be combined with the goals and the installation was not intended for technical laymen anyway, it was omitted. Both the v8 and the v9-1 were connected with reversed polarity due to a series of errors. In both cases, the HV fuse tripped via the body diode. In the v8, the precharge IGBT was also destroyed, but worked again after it was replaced. In the v9-1, the low-side FET was permanently conductive, but even after replacing it, no problems occurred in this context in the following months.

  Our risk assessment for the SSDV3 project expressly excluded an input fuse, since the risk of a false-trip during flight causing loss-of-signal and uncontrolled flyaway would be much more hazardous than a localized electrical fault. Nonetheless UBCO Aerospace has also had its own history of reverse polarity faults, so an RPP circuit was considered as the *minimum* protective circuitry to be added to the project.

- During the first tests, the 3 inch ground wire of an oscilloscope probe was accidentally shorted with the solder terminal of one of the 50 V ceramic input capacitors. Thankfully the overcurrent protection was active on the test bench supply at the time, and only a very brief spark occurred between the wire and the capacitor. This did cause minor charring on the capacitor, but we neglected it and testing continued. Two weeks later, the same module entered a short-circuit failure 4 minutes and 37 seconds into a thermal test. It took about a week to figure out the ceramic input capacitor had been damaged, and the thermal testing had caused internal cracks to worsen, leading to the corner of the capacitor to explode. The damage can be seen in Figure 7.1 and Figure 7.2. The module worked fine after replacing the damaged capacitor, but the whole ordeal just showed how sensitive ceramic capacitors are to fracture.

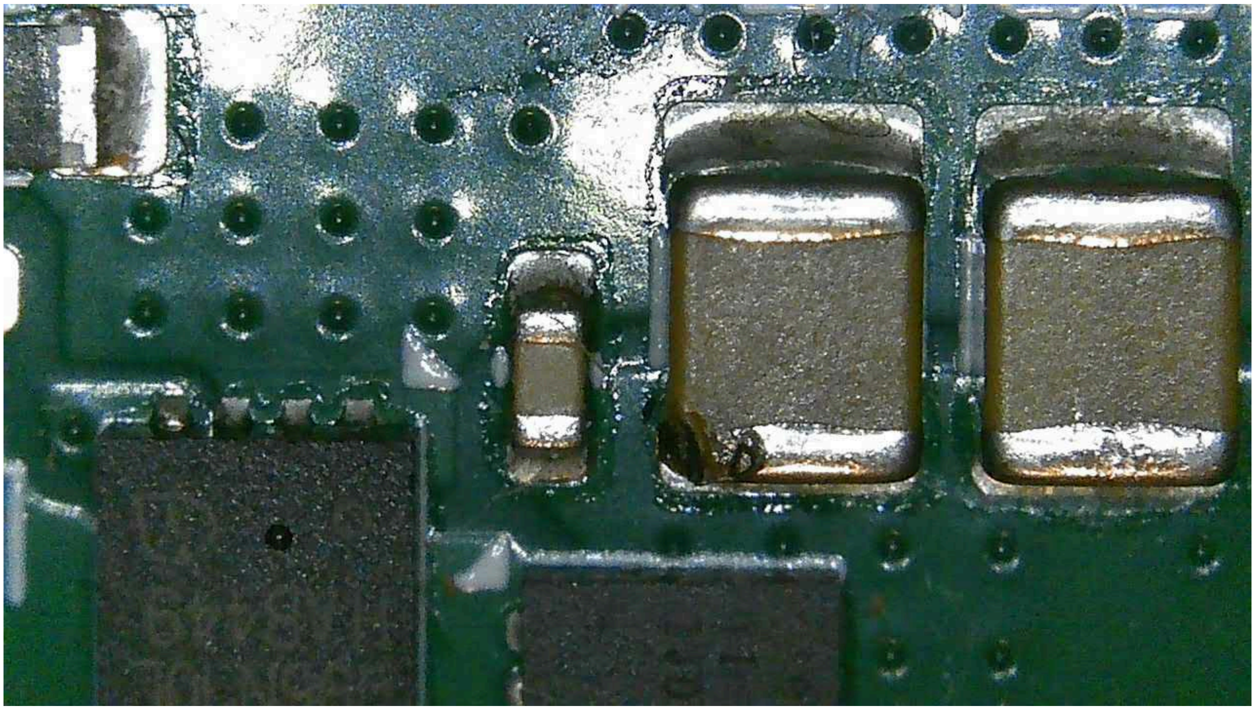


Figure 7.1: Microscope view of the damaged input capacitor.

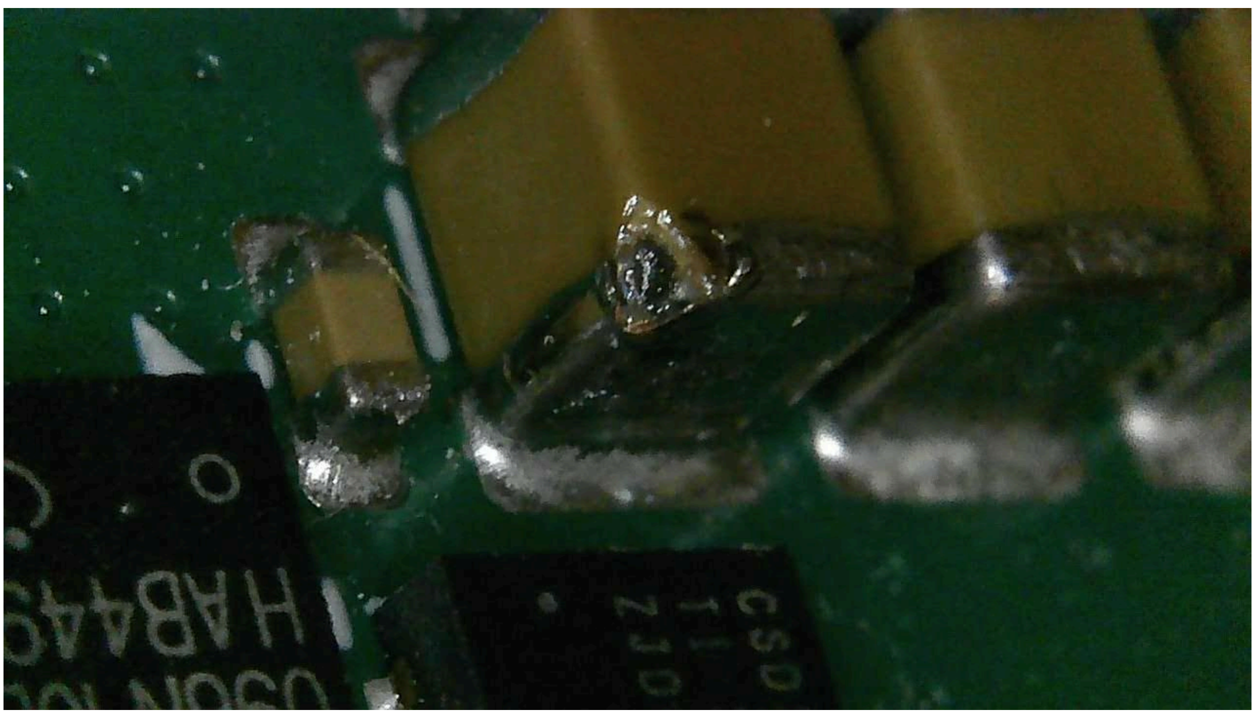


Figure 7.2: Microscope view of the damaged input capacitor from the side.

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## Further Reading

| Influential source material for this project | Notes |
|---|---|
| Logan Boone Projects \| Super Step Down V2 | *This is the predecessor to the Super Step Down V3 project.* |
| Rootthecause \| Open Source DC/DC Converter | *Inspired much of the design process and testing for the project.* |
| Jim Williams \| Switching Regulators for Poets - A Gentle Guide for the Trepidatious | *The app. note might be 40 years old, and some terms are outdated, but it's fun to read! Plus Appendix C was extremely useful during the design.* |
| **Reviews of generic COTS buck converters** | **Notes** |
| lygte-info.dk \| Performance test of Adjustable buck converter LM2596 | |
| lygte-info.dk \| Test of DC-DC 10A 6.5-60V to 1.25-30V | |
| lygte-info.dk \| Converter DC-DC 3A 4.75-23V to 1.0-17V | |